# How long a properly colored path must the union of two Hamiltonian paths contain?

Alejandro Lizardi*

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**Abstract**

Overlay two Hamiltonian paths on a common vertex set of size $n$, one blue and one red. How long a properly colored path – consecutive edges alternating in color – must the union contain? Every such union is $H_\sigma$ for a permutation $\sigma$ of $\{0, \ldots, n-1\}$, with blue edges joining consecutive positions and red edges consecutive values; write $\rho(\sigma)$ for the maximum number of vertices on a properly colored path in $H_\sigma$, and $\rho_{\min}(n) = \min_\sigma \rho(\sigma)$. We prove $\rho_{\min}(n) = O(\sqrt{n})$, explicitly $\rho_{\min}(n) \le 8\sqrt{n} + 20$ for $n \ge 100$, by inflating a gadget whose *transit number* equals 3 – and we prove that 3 is optimal for every gadget size. In the other direction we show that block statistics yield no lower bound: there are *block-free* permutations ($b(\sigma) = n$ maximal $\pm 1$-blocks) with $\rho = \Theta(\sqrt{n})$, so $\inf_\sigma \rho(\sigma)/b(\sigma) = 0$, and the inequality $\rho \ge b(\sigma)$, which holds exhaustively through $n = 12$, first fails at $n = 14$. The problem the paper poses is the lower bound: *does* $\rho_{\min}(n) \to \infty$*?* Two overlaid spanning paths look far too connected for the answer to be no, yet no lower bound growing with $n$ is known, and our results show that any such bound must stop at $O(\sqrt{n})$.

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## 1 Introduction

Let $S_n$ denote the set of bijections $\sigma\colon \{0, 1, \ldots, n-1\} \to \{0, 1, \ldots, n-1\}$. Associate with $\sigma \in S_n$ the two-edge-colored multigraph $H_\sigma$ on vertex set $\{0, 1, \ldots, n-1\}$ with

- **P-edges** ("position", blue): $\{i, i+1\}$ for $0 \le i \le n-2$;
- **V-edges** ("value", red): $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ for $0 \le v \le n-2$.

*Independent researcher, `alejlizardi05@gmail.com`. The proofs and computations in this paper were AI-assisted by Claude, under the Periapsis research structure. All computational claims are reproducible from the ancillary files accompanying this submission; see Appendix D.

Each color class is a Hamiltonian path. A vertex pair may carry both a P-edge and a V-edge; these are retained as *parallel edges of different colors* ("doubled pairs"). Conversely, the union of any two Hamiltonian paths on a common vertex set, with one path blue and the other red, is color-isomorphic to $H_\sigma$ for some $\sigma$ (identify the vertex set with $\{0, \ldots, n-1\}$ along the blue path, and let $\sigma$ rank the vertices along the red path).

A *properly colored path* (PC path) is a simple path—all vertices distinct—whose consecutive edges have different colors. A single vertex is a PC path. Let $\rho(\sigma)$ be the maximum number of vertices on a PC path in $H_\sigma$, and

$$\rho_{\min}(n) \;=\; \min_{\sigma \in S_n} \rho(\sigma).$$

**A useful picture.** Plot $\sigma$ as the point set $\{(i, \sigma(i)) : 0 \le i \le n-1\}$ in the plane. A P-edge joins horizontally adjacent columns; a V-edge joins vertically adjacent rows. A PC path traces an alternating horizontal/vertical staircase through this grid. Both color classes have maximum degree 2, so $H_\sigma$ has maximum degree 4; this places the problem outside the regimes of standard properly colored path results, which typically require minimum degree linear in $n$ [3].

### 1.1 Motivation: an extremal problem about alternating paths

The question is a natural extremal problem about alternating paths. Properly colored ("alternating") paths and cycles in edge-colored graphs have been studied extensively since the survey of Bang-Jensen and Gutin [1]; see also [2, 3]. The positive results in that line are sufficient conditions—typically large color degree or dense hosts—for long PC paths or cycles to exist. The graphs $H_\sigma$ sit at the opposite end of the spectrum: every vertex has at most two edges of each color, and the question is extremal rather than Dirac-type. We are aware of no prior work on the union of two Hamiltonian paths in this guise.

### 1.2 Results

The clean self-contained question is:

> *Does there exist $c > 0$ such that every two-edge-colored multigraph that is the union of two Hamiltonian paths on the same vertex set contains a properly colored simple path on at least $cn$ vertices?*

We answer it in the negative, and quantitatively.

**Theorem 1.1** (Main Theorem). *$\rho_{\min}(n) = O(\sqrt{n}\,)$. Explicitly, $\rho_{\min}(n) \le 8\sqrt{n} + 20$ for all $n \ge 100$.*

The question we consider central points the other way:

> *Does $\rho_{\min}(n) \to \infty$?*

A union of two Hamiltonian paths is spanning and connected, and every vertex sits on two long monochromatic paths; it is hard to believe that its longest properly colored path could stay *bounded* as $n \to \infty$. Nothing we know rules it out. The transit floor (Section 7) quantifies part of the difficulty: the construction machinery cannot be improved by better gadgets, and

Theorem 1.1 itself caps any future lower bound at $O(\sqrt{n})$. We hope the question finds attackers. Section 8 closes with three further open problems supported by exact data, among them a $k$-color analogue of the main question whose smallest counterexample – a single triple of orderings of eleven points – we exhibit.

The construction behind the main theorem is explicit. For even $b \geq 6$ let

$$T_b \;=\; (0,\ b{-}1,\ 2,\ 3,\ \ldots,\ b{-}2,\ 1) \in S_b$$

be the identity permutation with the values 1 and $b{-}1$ exchanged, and for $a \geq 2$ define $\sigma_{a,b} \in S_{ab}$ by

$$\sigma_{a,b}(jb + r) \;=\; jb + T_b(r) \qquad (0 \leq j < a,\ 0 \leq r < b),$$

i.e. $a$ position-consecutive blocks, each a copy of $T_b$, on consecutive value ranges. We prove the two-sided estimate

$$2a + 2b - 7 \;\leq\; \rho(\sigma_{a,b}) \;\leq\; 4a + 2b \qquad (a \geq 2,\ b \geq 6 \text{ even}), \tag{1}$$

the upper bound by the inflation calculus below (Theorem 6.1), the lower bound by an explicit path (Proposition 6.3). Thus $\rho(\sigma_{a,b}) = \Theta(a + b)$: with $a \asymp b \asymp \sqrt{n}$ the family realizes the order $\sqrt{n}$ exactly, so the upper bound is tight for this family and not an artifact of a construction whose true value is smaller. (For $a = 2$ both bounds in (1) are weak; the construction is informative for $a \geq 3$, and asymptotically for $a \to \infty$.) Exact machine computation gives $\rho(\sigma_{a,b}) = 2a + 2b - 4$ at every tested point up to $n = 720$ (Appendix D); we use this exact formula nowhere in the proofs and state it only as data.

The proof of the upper bounds has two independent components.

**1. An inflation calculus (Sections 3–4).** Call a set $B$ of consecutive positions whose $\sigma$-image is a set of consecutive values an *interval block*. A short argument (Lemma 3.2) shows that at most *four* edges of $H_\sigma$ join $B$ to its complement, and that they attach at four canonical *terminal* vertices of $B$, with prescribed colors. If $\sigma = \tau[\pi_1, \ldots, \pi_a]$ is built by inflation (each position block an interval block with pattern $\pi_i$, arranged by an outer permutation $\tau$), then any PC path decomposes into at most $2a - 1$ *visits* to blocks, and each visit other than the first and last must enter and leave through terminals subject to color constraints. Defining the *transit number* $M(\pi)$ as the largest number of vertices such a constrained passage (*admissible visit*) through $H_\pi$ can collect, we obtain (Theorem 4.4)

$$\rho\big(\tau[\pi_1, \ldots, \pi_a]\big) \;\leq\; (2a-3)\,\max_i M(\pi_i) \;+\; 2\,\max_i |\pi_i|.$$

**2. A classification of the gadget's admissible visits (Section 5).** For every even $b \geq 6$ we determine the admissible visits of $H_{T_b}$ *completely* (Theorem 5.4): up to reversal there are exactly five, of sizes $1, 2, 2, 3, 3$, so $M(T_b) = 3$, independent of $b$. The mechanism is a parity obstruction: $H_{T_b}$ contains a long run of doubled edges which any terminal-to-terminal passage must either avoid or cross completely, and a full crossing forces the color of the departing edge precisely so that, for even $b$, every continuation is blocked. For odd $b$ the parity reverses and a transit can collect all $b$ vertices. The even/odd contrast shows that the parity, not some feature of the argument, is what bounds the transit number.

Combining the two components and balancing $a$ against $b$ yields Theorem 1.1 (proved in Section 6). The two-visit refinement of the classification (Corollary 5.7) gives the stated constant

$8\sqrt{n}+20$; the value $M(T_b) = 3$ alone, without the full classification, already suffices for the weaker $\rho_{\min}(n) \le 10\sqrt{n}+10$ (Remark 6.2).

The paper then proves a calibration result.

**3. The transit floor (Section 7).** The inflation calculus makes one parameter decisive: how small a transit number a pattern can have. We prove that the gadgets above are optimal in this parameter:

$$\mu(b) \ := \ \min_{\pi \in S_b} M(\pi) \ = \ 3 \qquad \text{for every } b \ge 3.$$

The lower bound $M(\pi) \ge 3$ for *every* pattern is proved by an elementary matching-union argument: choosing one matching inside each color's Hamiltonian path with prescribed exposed vertices forces their union to contain a properly colored terminal-to-terminal path that is itself an admissible visit of size at least 3. The upper bound is $T_b$ for even $b$ together with a sibling gadget $T_b'$ (the identity with values 2 and $b-1$ exchanged) for odd $b$, whose admissible visits we also classify completely (Appendix A). Consequently no choice of inflation blocks can improve the visit-count bookkeeping of the method by more than constants.

Section 8 collects what remains of the lower-bound question: $\rho_{\min}(n) \to \infty$ is open, and by Theorem 1.1 a lower bound of order $\sqrt{n}$ would be best possible.

## 2 Preliminaries

Throughout, $\sigma \in S_n$ and $H_\sigma$ are as in Section 1; colors are P and V, and for a color $c$ we write $\bar{c}$ for the other color. Paths are sequences of distinct vertices $u_1, \ldots, u_k$ ($k \ge 1$) together with, for each consecutive pair, a chosen edge (with its color) joining them; in a doubled pair the path commits to one of the two parallel edges. A path is *properly colored* (PC) if consecutive edges have different colors. The *size* of a path is its number of vertices $k$; a PC path on $k$ vertices has $k-1$ edges. Since each vertex pair carries at most one edge of each color and paths have distinct vertices, a path uses pairwise distinct edges, even in the multigraph sense.

**Observation 2.1.** A two-edge-colored multigraph is the union of a blue and a red Hamiltonian path on a common vertex set of size $n$ if and only if it is color-isomorphic to $H_\sigma$ for some $\sigma \in S_n$. Hence Theorem 1.1 is equivalent to its abstract formulation.

**Remark 2.2** (Two notions of block)**.** Two block notions recur, and we keep them distinct. An *interval block* (Definition 3.1) is any set of consecutive positions whose values are consecutive; it carries an arbitrary *pattern*, and is the unit of the inflation calculus. A $\pm 1$-*block* (Definition 2.3) is the special case in which consecutive values differ by a fixed $\pm 1$; these are intrinsic to $\sigma$, partition the positions, and are the unit of the lower-bound discussion (Section 8). Every $\pm 1$-block is an interval block; the converse fails.

**Definition 2.3** ($\pm 1$-block)**.** A $\pm 1$-*block* of $\sigma$ is a maximal interval of positions $[p_\ell, p_r]$ such that either $p_\ell = p_r$ (a *singleton*) or there is a fixed $\varepsilon \in \{+1, -1\}$ with $\sigma(i+1) - \sigma(i) = \varepsilon$ for all $p_\ell \le i < p_r$. The $\pm 1$-blocks partition $\{0, \ldots, n-1\}$; write $b(\sigma)$ for their number and $s_{\max}(\sigma)$ for the largest size.

Every internal edge of a $\pm 1$-block is doubled (consecutive positions hold consecutive values), so a $\pm 1$-block can be traversed in order with freely chosen alternating colors. This single fact gives two unconditional bounds.

**Proposition 2.4** (Largest-block and one-vertex bounds)**.** *For every* $\sigma \in S_n$, $\rho(\sigma) \geq s_{\max}(\sigma)$*; and if* $b(\sigma) \geq 2$*, then* $\rho(\sigma) \geq s_{\max}(\sigma) + 1$*.*

*Proof.* Let $B$ be a $\pm 1$-block of size $s_{\max}$. Its internal edges are doubled, so traversing $B$ in order with alternating $\mathrm{P}, \mathrm{V}, \dots$ is a PC path on $s_{\max}$ vertices: this gives the first bound.

For the second, if $s_{\max} = 1$ then all blocks are singletons; with $b \geq 2$ the P-edge $\{0, 1\}$ exists and is a PC path on $2 = s_{\max} + 1$ vertices. Otherwise $s_{\max} \geq 2$. Since $b \geq 2$, a largest block $B$ has a position-adjacent $\pm 1$-block, hence an external edge $e$ of some color $c$ at a port $p \in \{p_\ell, p_r\}$ (Lemma 3.2 below gives the port structure for interval blocks, of which $B$ is one). Traverse $B$ from the port opposite $p$ toward $p$, choosing the starting color so that the last internal edge has color $\neq c$ (the doubled internal edges allow either parity); then $e$, of color $c$, may be appended at $p$, yielding a PC path on $s_{\max} + 1$ vertices. □

These are weak in absolute terms – the gadget family below has $s_{\max} = b$ fixed while $n \to \infty$ – but they are the floor any combined bound must clear, and they are the only unconditional lower bounds we have (Section 8).

## 3 Interval blocks and ports

We now develop the machinery for the main theorem. The unit is the interval block of Remark 2.2.

**Definition 3.1** (Interval block, pattern)**.** An *interval block* of $\sigma \in S_n$ is a set of positions $B = \{p_\ell, p_\ell + 1, \dots, p_r\}$ whose image is a set of consecutive values $\sigma(B) = \{w_\ell, \dots, w_r\}$. Its *pattern* is the permutation $\pi \in S_b$, $b = |B|$, defined by $\pi(t) = \sigma(p_\ell + t) - w_\ell$.

**Lemma 3.2** (Port structure)**.** *Let* $B$ *be an interval block of* $\sigma$ *with positions* $[p_\ell, p_r]$*, values* $[w_\ell, w_r]$*, and pattern* $\pi$*. Then:*

*(a) The sub-multigraph of* $H_\sigma$ *induced on* $B$ *is color-isomorphic to* $H_\pi$ *under the map* $p \mapsto p - p_\ell$*.*

*(b) Every edge of* $H_\sigma$ *with exactly one endpoint in* $B$ *is one of at most four: the P-edges* $\{p_\ell - 1, p_\ell\}$ *and* $\{p_r, p_r + 1\}$ *(when they exist), whose endpoints in* $B$ *are* $p_\ell$ *and* $p_r$*; and the V-edges* $\{\sigma^{-1}(w_\ell - 1), \sigma^{-1}(w_\ell)\}$ *and* $\{\sigma^{-1}(w_r), \sigma^{-1}(w_r + 1)\}$ *(when they exist), whose endpoints in* $B$ *are* $\sigma^{-1}(w_\ell)$ *and* $\sigma^{-1}(w_r)$*.*

*Proof.* (a) P-edges internal to $B$ are the pairs $\{p, p{+}1\} \subseteq [p_\ell, p_r]$, which correspond exactly to the P-edges of $H_\pi$. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v{+}1)\}$ has both endpoints in $B$ iff both $v, v{+}1 \in [w_\ell, w_r]$ (as $\sigma(B) = [w_\ell, w_r]$ and $\sigma$ is a bijection), and these correspond exactly to the V-edges of $H_\pi$ under the stated map, since $\pi^{-1}(v - w_\ell) = \sigma^{-1}(v) - p_\ell$ for $v \in [w_\ell, w_r]$. Doubled pairs match doubled pairs, so the isomorphism preserves colors.

(b) A P-edge $\{p, p + 1\}$ with exactly one endpoint in $[p_\ell, p_r]$ satisfies $p + 1 = p_\ell$ or $p = p_r$. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v + 1)\}$ with exactly one endpoint in $B$ has exactly one of $v, v + 1$ in $[w_\ell, w_r]$, forcing $v + 1 = w_\ell$ or $v = w_r$. □

**Definition 3.3** (Terminals, roles, port colors)**.** Let $\pi \in S_b$, $b \geq 2$. The *terminals* of $H_\pi$ are the four *roles*

$$p_0 = \text{vertex } 0, \quad p_1 = \text{vertex } b - 1, \quad v_0 = \text{vertex } \pi^{-1}(0), \quad v_1 = \text{vertex } \pi^{-1}(b - 1),$$

with *port colors* $\mathrm{pc}(p_0) = \mathrm{pc}(p_1) = \mathrm{P}$ and $\mathrm{pc}(v_0) = \mathrm{pc}(v_1) = \mathrm{V}$. Distinct roles may occupy the same vertex (a role is a label, not a vertex); since $b \geq 2$, two roles with the *same* port color always occupy distinct vertices.

By Lemma 3.2, in any interval block with pattern $\pi$ the at most four outgoing edges attach (inside the block) precisely at the terminals, and the color of an outgoing edge equals the port color of the role at which it attaches.

**Definition 3.4** (Admissible visit, transit number)**.** Let $\pi \in S_b$, $b \geq 2$. An *admissible visit* of $H_\pi$ is either

(i) a PC path $u_1, \ldots, u_k$ in $H_\pi$ with $k \geq 2$, together with a choice of roles $r$ at $u_1$ and $r'$ at $u_k$, such that the first edge has color $\neq \mathrm{pc}(r)$ and the last edge has color $\neq \mathrm{pc}(r')$; or

(ii) a single vertex carrying two distinct roles $r, r'$ with $\mathrm{pc}(r) \neq \mathrm{pc}(r')$.

The *transit number* $M(\pi)$ is the maximum size of an admissible visit, and $M(\pi) = 0$ if none exists. For $b \leq 1$ set $M(\pi) = 1$.

The color constraints encode that a visit is entered and left by external edges of the roles' port colors (which differ from the first and last internal edges by the PC condition). Two trivial but load-bearing facts:

**Lemma 3.5** (Reversal; endpoint constraint)**.** (a) *If $(u_1, \ldots, u_k; r, r')$ is an admissible visit, so is the reversed $(u_k, \ldots, u_1; r', r)$; admissible visits may thus be classified as undirected vertex sequences with unordered role pairs.* (b) *In a type-(i) admissible visit, both $u_1$ and $u_k$ are terminal vertices, the roles $r, r'$ are distinct, and the conditions read: first-edge color $= \overline{\mathrm{pc}(r)}$, last-edge color $= \overline{\mathrm{pc}(r')}$.*

*Proof.* (a) Reversal swaps first and last edges and the two endpoint constraints. (b) The roles occupy $u_1 \neq u_k$, hence are distinct; the rest is the definition together with the fact that with two colors "$\neq \mathrm{pc}(r)$" means "$= \overline{\mathrm{pc}(r)}$". □

# 4 Inflation and the inflation lemma

**Definition 4.1** (Inflation)**.** Let $a \geq 2$, let $\tau$ be a permutation of $\{0, \ldots, a-1\}$, and let $\pi_1, \ldots, \pi_a$ be permutations of sizes $b_1, \ldots, b_a \geq 2$, $N = \sum_i b_i$. Put $s_i = \sum_{j<i} b_j$ (position offsets) and $t_i = \sum_{j\colon\, \tau(j)<\tau(i)} b_j$ (value offsets). The *inflation* $\sigma = \tau[\pi_1, \ldots, \pi_a] \in S_N$ is

$$\sigma(s_i + r) = t_i + \pi_i(r), \qquad i = 1, \ldots, a, \quad 0 \leq r < b_i.$$

(All permutations in this paper act on $\{0, \ldots, \mathrm{size} - 1\}$; the block subscripts $\pi_1, \ldots, \pi_a$ and the offset indices $i, j$ enumerate the $a$ blocks and range over $1, \ldots, a$, a labelling of blocks rather than a permutation domain.)

The value intervals $[t_i, t_i + b_i - 1]$ partition $\{0, \ldots, N-1\}$, so $\sigma$ is a bijection. Each position block $B_i = [s_i, s_i + b_i - 1]$ is an interval block of $\sigma$ with value interval $[t_i, t_i + b_i - 1]$ and pattern $\pi_i$, so Lemma 3.2 applies to all blocks simultaneously. Edges of $H_\sigma$ with endpoints in different blocks are *cross edges*.

**Lemma 4.2.** *$H_{\tau[\pi_1,\ldots,\pi_a]}$ has exactly $2(a-1)$ cross edges, counted with multiplicity: $a-1$ P-colored (one per consecutive pair of position blocks) and $a-1$ V-colored (one per consecutive pair of value intervals). Each block is incident to at most 4 cross edges, attaching at its terminals as in Lemma 3.2(b), and each terminal role of each block carries at most one cross edge.*

*Proof.* A P-edge $\{p, p+1\}$ is a cross edge iff $p+1 \in \{s_2, \ldots, s_a\}$: $a-1$ edges. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ is a cross edge iff $v, v+1$ lie in different value intervals, which happens for exactly the $a-1$ values $v$ ending a value interval. The attachment statement is Lemma 3.2(b); per block, each of the four port positions is met by at most one cross edge. □

**Lemma 4.3** (Visit decomposition)**.** *Let $\sigma = \tau[\pi_1, \ldots, \pi_a]$, $a \geq 2$, and let $Q$ be a PC path in $H_\sigma$. Partition the vertex sequence of $Q$ into maximal runs of $Q$-consecutive vertices lying in a common block; call these runs* visits*. Then:*

- *(a) Consecutive visits are joined by exactly one cross edge; distinct consecutive pairs use distinct cross edges. Hence if $Q$ has $q$ visits, then $q-1 \leq 2(a-1)$, i.e. $q \leq 2a-1$.*
- *(b) Every visit that is neither the first nor the last is an admissible visit of the copy of $H_{\pi_i}$ on its block; in particular it has at most $M(\pi_i)$ vertices.*
- *(c) The first and the last visit have at most $\max_i b_i$ vertices each.*

*Proof.* (a) The edge of $Q$ joining the last vertex of one visit to the first vertex of the next has endpoints in different blocks (maximality of runs), so it is a cross edge. $Q$ has distinct vertices, hence pairwise distinct edges, so the $q-1$ connecting edges are distinct cross edges; there are $2(a-1)$ of them in all (Lemma 4.2).

(b) Let $W = u_1, \ldots, u_k$ be an interior visit in block $B_i$. There is an *entry* cross edge $e_{\text{in}}$ of $Q$ at $u_1$ and an *exit* cross edge $e_{\text{out}}$ at $u_k$. By Lemma 4.2, $e_{\text{in}}$ attaches at a terminal role $r$ at $u_1$ with color $\text{pc}(r)$, and $e_{\text{out}}$ at a role $r'$ at $u_k$ with color $\text{pc}(r')$. All edges of $Q$ between vertices of $W$ lie in $B_i$, so by Lemma 3.2(a) $W$ is a PC path in $H_{\pi_i}$. If $k \geq 2$: the PC condition at $u_1$ (between $e_{\text{in}}$ and the first edge of $W$) forces the first edge of $W$ to have color $\neq \text{pc}(r)$, similarly at $u_k$; so $W$ with roles $r, r'$ is admissible of type (i). If $k = 1$: the PC condition at $u_1$ between $e_{\text{in}}$ and $e_{\text{out}}$ forces $\text{pc}(r) \neq \text{pc}(r')$: admissible of type (ii).

(c) A visit lies in one block. □

**Theorem 4.4** (Inflation Lemma)**.** *For $a \geq 2$,*

$$\rho\big(\tau[\pi_1, \ldots, \pi_a]\big) \;\leq\; (2a-3)\max_i M(\pi_i) \;+\; 2\max_i b_i.$$

*Proof.* Let $Q$ attain $\rho$ with $q$ visits. If $q = 1$, then $\rho \leq \max_i b_i$. If $q \geq 2$, then by Lemma 4.3 the $q-2 \leq 2a-3$ interior visits have at most $\max_i M(\pi_i)$ vertices each and the two end visits at most $\max_i b_i$ each; visits partition the vertices of $Q$. □

Theorem 4.4 holds for *every* outer permutation $\tau$. The problem is thereby reduced to finding patterns with small transit number relative to their size.

## 5 The gadget: classification of admissible visits of $H_{T_b}$

**Definition 5.1.** For even $b \geq 6$ let $T_b \in S_b$ be given by

$$T_b(0) = 0, \qquad T_b(1) = b-1, \qquad T_b(i) = i \ \ (2 \leq i \leq b-2), \qquad T_b(b-1) = 1.$$

Thus $T_b^{-1}$ fixes 0 and each $v \in [2, b-2]$ and swaps $1 \leftrightarrow b-1$ ($T_b$ is an involution). Recall $b \geq 6$.

**Lemma 5.2** (Inventory)**.** *The edges of $H_{T_b}$ are exactly:*

- doubled pairs $\{i, i+1\}$ *for* $2 \leq i \leq b-3$ *(each carrying one P- and one V-edge);*
- P-only edges $\{0,1\}$, $\{1,2\}$, $\{b-2, b-1\}$*;*
- V-only edges $\{0, b-1\}$, $\{2, b-1\}$, $\{1, b-2\}$.

*The full edge lists at the relevant vertices are:*

$$\begin{array}{ll}
\textit{vertex } 0: & \mathrm{P}\,\{0,1\};\ \ \mathrm{V}\,\{0,b{-}1\} \quad (\textit{degree } 2),\\
\textit{vertex } 1: & \mathrm{P}\,\{0,1\},\ \mathrm{P}\,\{1,2\};\ \ \mathrm{V}\,\{1,b{-}2\} \quad (\textit{unique V-edge}),\\
\textit{vertex } b{-}1: & \mathrm{P}\,\{b{-}2,b{-}1\};\ \ \mathrm{V}\,\{0,b{-}1\},\ \mathrm{V}\,\{2,b{-}1\} \quad (\textit{unique P-edge}),\\
\textit{vertex } 2: & \mathrm{P}\,\{1,2\};\ \textit{doubled } \{2,3\};\ \mathrm{V}\,\{2,b{-}1\},\\
\textit{vertex } b{-}2: & \textit{doubled } \{b{-}3,b{-}2\};\ \mathrm{P}\,\{b{-}2,b{-}1\};\ \mathrm{V}\,\{1,b{-}2\},\\
\textit{vertex } i,\ 3 \leq i \leq b-3: & \textit{doubled } \{i{-}1,i\} \textit{ and } \{i,i{+}1\} \textit{ only.}
\end{array}$$

*The terminals are $p_0 = v_0 =$ vertex 0, $v_1 =$ vertex 1, $p_1 =$ vertex $b-1$.*

*Proof.* P-edges are $\{i, i+1\}$, $0 \leq i \leq b-2$. The V-edges, indexed by $v = 0, \ldots, b-2$, are $\{T_b^{-1}(v), T_b^{-1}(v+1)\}$:

$$v=0:\ \{0,b{-}1\}; \qquad v=1:\ \{b{-}1,2\}; \qquad 2 \leq v \leq b-3:\ \{v,v{+}1\}; \qquad v=b{-}2:\ \{b{-}2,1\}.$$

A pair $\{i, i+1\}$ is doubled iff it appears in both lists: exactly for $2 \leq i \leq b-3$. For $b \geq 6$: $\{0,1\}$ doubled would need $b-1=1$; $\{1,2\}$ would need $b-3=1$; $\{b-2,b-1\}$ would need $b-3=1$; all false, and the three long V-edges join non-adjacent positions since $b-1, b-3 \geq 3$. The per-vertex lists and the terminals ($T_b^{-1}(0) = 0$, $T_b^{-1}(b-1) = 1$) follow by inspection. $\square$

Let $S = \{2, 3, \ldots, b-2\}$ (the *segment*), $|S| = b-3$. By Lemma 5.2, the sub-multigraph on $S$ is the doubled path $2 - 3 - \cdots - (b-2)$; the only edges from $S$ to its complement are the two at vertex 2 (P $\{1,2\}$, V $\{2, b-1\}$) and the two at vertex $b-2$ (P $\{b-2, b-1\}$, V $\{1, b-2\}$); and no terminal lies in $S$. Figure 1 pictures the inventory.

**Lemma 5.3** (Segment behavior)**.** *Let $Q$ be a PC path in $H_{T_b}$.*

*(a) While inside $S$, $Q$ moves monotonically in position (each interior vertex of $S$ is adjacent only to its two position-neighbors), and $Q$ can leave $S$ only from vertex 2 or vertex $b-2$.*

*(b) Suppose $Q$ enters $S$ at one end (2 or $b-2$) by an edge of color $c$, at a moment when no vertex of $S$ has yet been visited, and later leaves $S$. Then either it leaves immediately from the entry vertex (using the other external edge there, of color $\bar{c}$), or it traverses all of $S$ to the far end; in the latter case its $b-4$ internal edges alternate colors starting with $\bar{c}$, so (since $b$ is even, making $b-4$ even) the last internal edge has color $c$, and any edge by which $Q$ then leaves $S$ has color $\bar{c}$.*

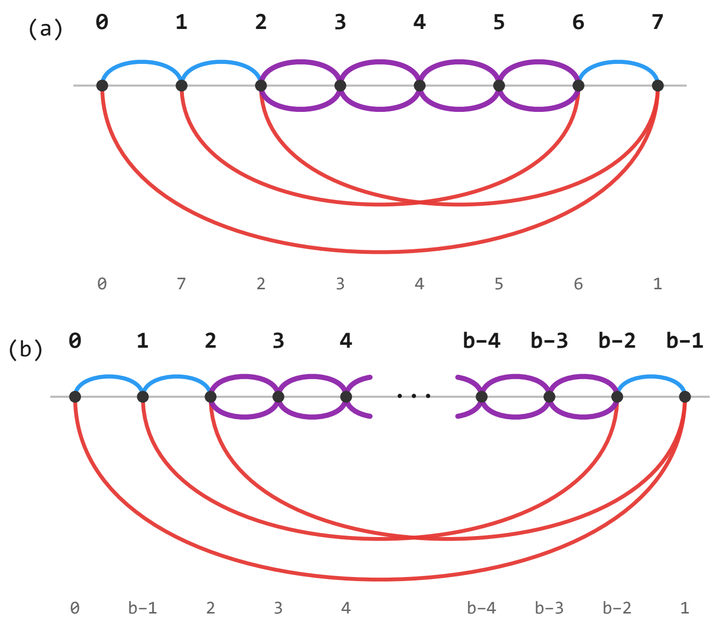


Figure 1: The gadget $H_{T_b}$: (a) the instance $T_8 = (0, 7, 2, 3, 4, 5, 6, 1)$; (b) general even $b$, the middle of the segment elided. Vertices are positions on a line, with the value $T_b(i)$ in small gray below; blue arcs above the line are P-edges, red arcs below are V-edges, and a purple pair of mirrored arcs is a doubled pair. The doubled run on the segment $S = \{2, \dots, b-2\}$ is flanked by the three P-only edges $\{0, 1\}, \{1, 2\}, \{b-2, b-1\}$ and the three V-only edges $\{0, b-1\}, \{2, b-1\}, \{1, b-2\}$; the terminals are 0 ($= p_0 = v_0$), 1 ($= v_1$), and $b-1$ ($= p_1$).

*Proof.* (a) is immediate from the inventory: a simple path cannot reverse inside a path graph. For (b): leaving from the entry vertex means the run in $S$ is that single vertex; the two external edges at 2 (resp. $b-2$) have different colors, and the PC condition forces the exit edge to differ from the entry edge, i.e. to be the other one, of color $\bar{c}$. Otherwise the run proceeds monotonically and exits only at the far end, having covered all of $S$ with internal edges $g_1, \dots, g_{b-4}$. The PC condition at the entry vertex gives $g_1 = \bar{c}$ (both colors are available on every doubled pair; the PC condition forces the alternation), alternation gives $g_m = \bar{c}$ for odd $m$ and $g_m = c$ for even $m$, and $b-4$ is even, so $g_{b-4} = c$; the PC condition at the far end forces the exit edge to have color $\neq c$. $\square$

## 5.1 The classification theorem

**Theorem 5.4** (Classification of admissible visits)**.** *Let $b \geq 6$ be even. Up to the reversal symmetry of Lemma 3.5, the graph $H_{T_b}$ has exactly five admissible visits. They are the following, each with the role pair shown:*

| | *vertices* | *edges (colors)* | *role pair* |
|---|---|---|---|
| *(V1)* | $0$ | — | $\{p_0, v_0\}$ |
| *(V2)* | $0,\ b-1$ | $\mathrm{V}\{0, b{-}1\}$ | $\{p_0, p_1\}$ |
| *(V3)* | $0,\ 1$ | $\mathrm{P}\{0, 1\}$ | $\{v_0, v_1\}$ |
| *(V4)* | $b-1,\ 0,\ 1$ | $\mathrm{V}\{0, b{-}1\},\ \mathrm{P}\{0, 1\}$ | $\{p_1, v_1\}$ |
| *(V5)* | $b-1,\ 2,\ 1$ | $\mathrm{V}\{2, b{-}1\},\ \mathrm{P}\{1, 2\}$ | $\{p_1, v_1\}$ |

*In particular: the role pairs $\{p_0, v_1\}$ and $\{v_0, p_1\}$ admit no visit; the only visit for $\{p_0, p_1\}$ is (V2); the only visit for $\{v_0, v_1\}$ is (V3); and $M(T_b) = 3$.*

*Proof.* That each listed visit is admissible is a direct check from Lemma 5.2. (V1): vertex 0 carries $p_0$ and $v_0$ with different port colors. (V2): first = last edge $\mathrm{V} \neq \mathrm{pc}(p_0) = \mathrm{pc}(p_1) = \mathrm{P}$. (V3): first = last edge $\mathrm{P} \neq \mathrm{pc}(v_0) = \mathrm{pc}(v_1) = \mathrm{V}$. (V4), (V5): first edge $\mathrm{V} \neq \mathrm{pc}(p_1)$, last edge $\mathrm{P} \neq \mathrm{pc}(v_1)$, and the middle vertex alternates the colors. The role pairs are forced: e.g. in (V2) the first edge is V, so the role at 0 has port color $\neq \mathrm{V}$, i.e. $p_0$; the same one-line check fixes each role pair.

It remains to show no other admissible visit exists. Type-(ii) visits need a vertex carrying two roles of different port colors; by Lemma 5.2 the only such vertex is $0 = p_0 = v_0$, giving (V1). A type-(i) visit starts at a terminal vertex with a constrained first color (Lemma 3.5(b)). The terminal vertices are $0, 1, b-1$, giving exactly four (start vertex, first color) *stems*:

(A) $(0, \mathrm{V})$ [role $p_0$], (B) $(0, \mathrm{P})$ [role $v_0$], (C) $(1, \mathrm{P})$ [role $v_1$], (D) $(b-1, \mathrm{V})$ [role $p_1$].

Every admissible visit, read from its first vertex, extends one of these stems, so it suffices to enumerate, for each stem, every PC continuation, recording the moments at which the path stands at a terminal vertex with a last-edge color admissible for a role there. We use the *forcing facts* from Lemma 5.2: (F1) vertex 0 has exactly one P-edge ($\{0, 1\}$) and one V-edge ($\{0, b-1\}$); (F2) vertex 1 has exactly one V-edge ($\{1, b-2\}$); (F3) vertex $b-1$ has exactly one P-edge ($\{b-2, b-1\}$); plus the segment behavior of Lemma 5.3. No vertex of $S$ is a terminal, so a path stranded in $S$ can never end an admissible visit there.

**Stem (A): start** $0$**, first edge** V**.** By (F1) the first edge is $\mathrm{V}\{0, b-1\}$; the path stands at $b-1$, last color V. *Record:* role $p_1$ at $b-1$ has $\mathrm{pc} = \mathrm{P} \neq \mathrm{V}$, so $(0, b-1)$ is admissible: this is (V2). Continue: next $\neq \mathrm{V}$, so by (F3) $\mathrm{P}\{b-2, b-1\}$; enter $S$ at $b-2$, entry color P, $S$ untouched. By Lemma 5.3(b):

- *Immediate exit at* $b-2$ by $\mathrm{V}\{1, b-2\}$: stand at 1, last color V; the only role at 1 is $v_1$ ($\mathrm{pc} = \mathrm{V}$): not admissible. Continue from 1: next $\neq \mathrm{V}$, so a P-edge; $\{0, 1\}$ excluded (0 visited), leaving $\mathrm{P}\{1, 2\}$ into $S$ at 2. From 2, next $\neq \mathrm{P}$: $\mathrm{V}\{2, b{-}1\}$ excluded ($b{-}1$ visited), leaving the V-side of $\{2, 3\}$; from 3 the path moves monotonically up and strands (exits are 2, behind it, and $b-2$, visited). No admissible visit.
- *Full crossing to* 2: arrival at 2 by color P (Lemma 5.3(b) with $c = \mathrm{P}$); exit must be V; the only V-external edge at 2 is $\{2, b-1\}$ with $b-1$ visited. Strands. No admissible visit.

*Stem (A) yields exactly (V2).*

**Stem (B): start** $0$**, first edge** P**.** By (F1) the first edge is $\mathrm{P}\{0, 1\}$; stand at 1, last color P. *Record:* role $v_1$ at 1 has $\mathrm{pc} = \mathrm{V} \neq \mathrm{P}$: admissible: (V3). Continue: next $\neq \mathrm{P}$, so by (F2) $\mathrm{V}\{1, b-2\}$; enter $S$ at $b-2$, entry color V, $S$ untouched. By Lemma 5.3(b):

- *Immediate exit at* $b-2$ by P$\{b-2, b-1\}$: stand at $b-1$, last color P; the only role at $b-1$ is $p_1$ (pc = P): not admissible. Continue from $b-1$: next $\neq$ P, so a V-edge; $\{0, b-1\}$ excluded (0 visited), leaving V$\{2, b-1\}$ into $S$ at 2. From 2, next $\neq$ V: P$\{1,2\}$ excluded (1 visited), leaving the P-side of $\{2,3\}$; monotone up and strands ($b-2$ visited). No admissible visit.

- *Full crossing to* 2: arrival color V; exit must be P; only P-external edge at 2 is $\{1,2\}$, and 1 is visited. Strands. No admissible visit.

*Stem (B) yields exactly (V3).*

**Stem (C): start** 1**, first edge** P. The P-edges at 1 are $\{0,1\}$ and $\{1,2\}$; two branches.

- *Branch* $1 \to 0$. At 0, last color P. *Record:* $v_0$ (pc = V) admissible: (V3) reversed. Continue: next $\neq$ P, by (F1) V$\{0, b-1\}$; at $b-1$, last color V. *Record:* $p_1$ admissible: (V4) reversed. Continue: next $\neq$ V, by (F3) P$\{b-2, b-1\}$ into $S$ at $b-2$, entry color P, $S$ untouched. Immediate exit by V$\{1, b-2\}$: 1 visited. Full crossing: arrival at 2 color P, exit must be V: only $\{2, b-1\}$, visited. Strands; branch closed.

- *Branch* $1 \to 2$. Enter $S$ at 2 by P$\{1,2\}$, entry color P, $S$ untouched. Immediate exit by V$\{2, b-1\}$: at $b-1$, last color V. *Record:* $p_1$ admissible: (V5) reversed. Continue from $b-1$: next $\neq$ V, by (F3) P$\{b-2, b-1\}$ into $S$ at $b-2$ – but $S$ is now touched (vertex 2): from $b-2$ the non-P continuations are V$\{1, b-2\}$ (1 visited) and the V-side of $\{b-3, b-2\}$, after which the path moves monotonically down and strands at 3 (2 visited). Branch closed. Alternatively, full crossing $2 \to b-2$ with entry color P arrives at $b-2$ by color P; exit must be V: only $\{1, b-2\}$, 1 visited. Strands; branch closed.

*Stem (C) yields exactly (V3), (V4), (V5) (reversed).*

**Stem (D): start** $b-1$**, first edge** V. The V-edges at $b-1$ are $\{0, b-1\}$ and $\{2, b-1\}$; two branches.

- *Branch* $b-1 \to 0$. At 0, last color V. *Record:* $p_0$ admissible: (V2) reversed. Continue: next $\neq$ V, by (F1) P$\{0,1\}$; at 1, last color P. *Record:* $v_1$ admissible: (V4). Continue: next $\neq$ P, by (F2) V$\{1, b-2\}$ into $S$ at $b-2$, entry color V, $S$ untouched. Immediate exit by P$\{b-2, b-1\}$: $b-1$ visited. Full crossing: arrival at 2 color V, exit must be P: only $\{1,2\}$, 1 visited. Strands; branch closed.

- *Branch* $b-1 \to 2$. Enter $S$ at 2 by V$\{2, b-1\}$, entry color V, $S$ untouched. Immediate exit by P$\{1,2\}$: at 1, last color P. *Record:* $v_1$ admissible: (V5). Continue from 1: next $\neq$ P, by (F2) V$\{1, b-2\}$ into $S$ at $b-2$ – $S$ touched (vertex 2): from $b-2$ the non-V continuations are P$\{b-2, b-1\}$ ($b-1$ visited) and the P-side of $\{b-3, b-2\}$, then monotone descent stranding at 3 (2 visited). Branch closed. Alternatively, full crossing $2 \to b-2$ with entry color V arrives by color V; exit must be P: only $\{b-2, b-1\}$, $b-1$ visited. Strands; branch closed.

*Stem (D) yields exactly (V2), (V4), (V5).*

Collecting the four stems: every admissible visit is one of (V1)–(V5) up to reversal, each realized only with the listed role pair (checked at each *Record* step). No admissible visit realizes $\{p_0, v_1\}$ or $\{v_0, p_1\}$, and the maximum size is 3, attained by (V4) and (V5). □

**Corollary 5.5.** *$M(T_b) = 3$ for every even $b \geq 6$, and $c^* := \liminf_n \rho_{\min}(n)/n$ satisfies $c^* \leq 6/b$ for every even $b \geq 6$; hence $c^* = 0$.*

*Proof.* $M(T_b) = 3$ is part of Theorem 5.4. Taking $\pi_i = T_b$ for all $i$ in Theorem 4.4 and letting $a \to \infty$: $\rho_{\min}(ab)/(ab) \leq \big((2a-3)\cdot 3 + 2b\big)/(ab) \to 6/b$. □

**Remark 5.6** (The parity is essential)**.** For *odd* $b$, $b-4$ is odd, so the last internal edge of a full crossing has color $\bar{c}$ and the exit edge may have color $c$. The route $0 \xrightarrow{\mathrm{P}} 1 \xrightarrow{\mathrm{V}} b-2 \rightsquigarrow 2 \xrightarrow{\mathrm{V}\{2,b-1\}} b-1$ (full crossing in the middle) then becomes properly colored, an admissible visit for $\{v_0, p_1\}$ of size $b$.

A sibling gadget $T'_b$ – the identity with the values 2 and $b-1$ exchanged – serves *odd* $b$ by the same mechanism with the good parity reversed; its admissible visits are classified completely in Appendix A, and it enters the transit-floor bound through Proposition 7.2.

**Corollary 5.7** (Two visits to one $T_b$-block)**.** *Let $\sigma = \tau[\pi_1, \ldots, \pi_a]$ be an inflation, $B_i$ a block whose pattern $\pi_i$ equals $T_b$ for some even $b \geq 6$, and $Q$ a PC path in $H_\sigma$. Then the interior visits of $Q$ to $B_i$ have total size at most 4.*

*Proof.* By Lemma 4.2, $B_i$ has at most four incident cross edges, one per role; each interior visit uses two distinct ones (entry and exit), and every cross edge is used at most once. Hence there are at most two interior visits; if at most one, its size is at most $M(T_b) = 3 \leq 4$. If two, they use all four cross edges, so their role pairs partition $\{p_0, p_1, v_0, v_1\}$, and they are vertex-disjoint (disjoint runs of the simple path $Q$). By Theorem 5.4 the three partitions give:

- $\{p_0, p_1\}/\{v_0, v_1\}$: the only visits are (V2)$= (0, b-1)$ and (V3)$= (0, 1)$, which share vertex 0 – impossible;
- $\{p_0, v_1\}/\{v_0, p_1\}$: no visits exist for either pair – impossible;
- $\{p_0, v_0\}/\{p_1, v_1\}$: the $\{p_0, v_0\}$ visit is (V1) at vertex 0, and the $\{p_1, v_1\}$ visit is (V4) or (V5); since (V4)$= (b-1, 0, 1)$ contains vertex 0 it cannot be disjoint from (V1), so the disjoint pair is (V1) with (V5), of sizes 1 and 3: total 4. (In every case the total is at most 4.)

□

# 6 Proof of the main theorem

We prove the bound advertised in Theorem 1.1. The family is $\sigma_{a,b} = \mathrm{id}_a[T_b, \ldots, T_b]$, i.e. $\sigma_{a,b}(jb+r) = jb + T_b(r)$, and the proof combines the two-visit bound for one $T_b$-block (Corollary 5.7) with the inflation count of Section 4.

**Theorem 6.1** (Family bound)**.** *For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \leq 4a + 2b$.*

*Proof.* Let $Q$ be a PC path in $H_{\sigma_{a,b}}$ with $q$ visits. If $q = 1$ then $|Q| \leq b$. Otherwise, by Lemma 4.3 the first and last visits have at most $b$ vertices each, and by Corollary 5.7 the interior visits contribute at most 4 per block, i.e. at most $4a$ in total. Hence $\rho(\sigma_{a,b}) \leq 4a + 2b$. □

*Proof of Theorem 1.1.* Let $b = 2\lfloor\sqrt{n}/2\rfloor$, so $b$ is even, $\sqrt{n} - 2 < b \leq \sqrt{n}$, and $b \geq 10$ since $n \geq 100$. Let $a = \lfloor n/b \rfloor \geq 2$ and $s = n - (a-1)b$, so $b \leq s < 2b$. Define $\sigma \in S_n$ as the inflation (identity outer permutation) of:

- if $s$ even: $a-1$ copies of $T_b$ and one copy of $T_s$ ($s$ even, $s \geq b \geq 10$);
- if $s$ odd: $a-1$ copies of $T_b$, one copy of $T_{s-3}$, and one block of size 3 with identity pattern $\iota_3 = (0,1,2)$ ($s-3$ even, $s-3 \geq b-2 \geq 8$, since $s$ odd and $b$ even force $s \geq b+1$).

The number of blocks is at most $a+1 \leq n/b+1$, and each block is one of $T_b$, $T_s$, $T_{s-3}$, or $\iota_3$. Interior visits contribute at most 4 per $T$-block (Corollary 5.7, which applies verbatim to $T_b$, $T_s$, $T_{s-3}$) and at most $2 \cdot 3 = 6$ for the single $\iota_3$-block (at most two interior visits, by Lemma 4.2, of size at most 3 each). The two end visits contribute at most $2(2b-1)$, since the largest block has size at most $2b-1$. Hence

$$\rho_{\min}(n) \;\leq\; 4\Big(\frac{n}{b}+1\Big)+6+2(2b-1) \;=\; \frac{4n}{b}+4b+8 \;\leq\; \big(4\sqrt{n}+10\big)+4\sqrt{n}+8 \;\leq\; 8\sqrt{n}+20,$$

using $4n/(\sqrt{n}-2) = 4\sqrt{n}+8\sqrt{n}/(\sqrt{n}-2) \leq 4\sqrt{n}+10$ for $\sqrt{n} \geq 10$. □

**Remark 6.2** (A weaker bound without the full classification)**.** If one uses only the value $M(T_b) = 3$ and not the two-visit refinement, the Inflation Lemma (Theorem 4.4) already gives $\rho(\sigma_{a,b}) \leq 6a+2b$ and, by the same choice of blocks (each of transit number at most 3), $\rho_{\min}(n) \leq (2m-3)\cdot 3 + 2(2b-1) \leq 6n/b + 4b - 5 \leq 10\sqrt{n}+10$ for $n \geq 100$. The refinement to $8\sqrt{n}+20$ is what the two-visit corollary buys.

**Proposition 6.3** (Explicit long path: the family is $\Theta(a+b)$)**.** *For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \geq 2a+2b-7$.*

*Proof.* Write $(j,\ell)$ for the vertex at position $jb+\ell$ ($0 \leq j < a$, $0 \leq \ell < b$), so block $j$ is a copy of $H_{T_b}$ on $\{(j,0),\ldots,(j,b-1)\}$, with values $(j,0) \mapsto jb$, $(j,1) \mapsto jb+b-1$, $(j,\ell) \mapsto jb+\ell$ for $2 \leq \ell \leq b-2$, and $(j,b-1) \mapsto jb+1$. Besides the internal edges (Lemma 5.2 in each block), the cross edges are the P-edges $\{(j,b-1),(j+1,0)\}$ and the V-edges $\{(j,1),(j+1,0)\}$. Within block $j$ note the V-edges $\{(j,0),(j,b-1)\}$ and $\{(j,2),(j,b-1)\}$.

The following walk is a PC path; colors are indicated over each step, and each edge exists by the inventory.

- *Stage 1 (block 0).* Start at $(0,b-2)$ and descend the segment to $(0,2)$ along the doubled run, choosing colors V, P, V, …; this uses $b-4$ edges, and since $b-4$ is even the last (into $(0,2)$) has color P. Then take V $\{(0,2),(0,b-1)\}$, then the P-cross edge to $(1,0)$.
- *Stage 2 (blocks $1,\ldots,a-2$).* From $(j,0)$ (entered by a P-cross edge) take V $\{(j,0),(j,b-1)\}$, then the P-cross edge to $(j+1,0)$; repeat.
- *Stage 3 (block $a-1$).* From $(a-1,0)$ (entered by a P-cross edge) take V $\{(a-1,0),(a-1,b-1)\}$, then P $\{(a-1,b-2),(a-1,b-1)\}$, then descend the segment from $(a-1,b-2)$ to $(a-1,2)$ choosing colors V, P, …, stopping at $(a-1,2)$.

All vertices are distinct (stages use disjoint blocks; within a stage the listed vertices are distinct), and consecutive edge colors differ at every junction. The vertex count is $(b-2)$ in stage 1 ($b-3$ segment vertices plus $(0,b-1)$), $2(a-2)$ in stage 2, and $(b-1)$ in stage 3 ($(a-1,0)$, $(a-1,b-1)$, and the $b-3$ segment vertices), totalling $2a+2b-7$. □

**Remark 6.4** (Constants; exact values)**.** Together, Theorem 6.1 and Proposition 6.3 give $2(a+b)-7 \leq \rho(\sigma_{a,b}) \leq 4a+2b$: the family is $\Theta(a+b)$, and with $a \asymp b$ realizes $\Theta(\sqrt{n})$, so the upper bound of Theorem 1.1 is achieved by this family up to the constant. Exact machine computation (Appendix D) gives $\rho(\sigma_{a,b}) = 2a+2b-4$ at every tested point up to $n = 720$; we conjecture this holds for all $a \geq 3$, even $b \geq 6$, but we neither prove nor use it. Optimizing the proven upper bound $4a + 2b$ at $n = ab$ with $b \asymp \sqrt{2n}$ gives $\rho_{\min}(n) \leq 4\sqrt{2n} + O(1) \approx 5.66\sqrt{n}$ along such $n$.

# 7 The transit floor: no pattern has transit number below three

The main theorem rests on patterns whose transit number is 3 while their size grows. Within the inflation calculus, one parameter is decisive: how small the transit number of a pattern can be made. Define

$$\mu(b) \;:=\; \min_{\pi \in S_b} M(\pi) \qquad (b \geq 2).$$

Any family of patterns with $\mu$-optimal transit numbers gives the best interior-visit bookkeeping the Inflation Lemma can produce. This section determines $\mu$ completely.

**Theorem 7.1** (Transit floor)**.** *$\mu(b) = 3$ for every $b \geq 3$. (For $b = 2$, both permutations have $M = 2$, so $\mu(2) = 2$.)*

The upper bound is a construction, the lower bound a structure theorem; we prove them in turn. Together they say the gadgets of this paper are optimal in the transit parameter: no choice of inflation blocks can beat $M = 3$ at any size.

## 7.1 The upper bound: gadgets at every size

**Proposition 7.2.** *$\mu(b) \leq 3$ for every $b \geq 3$.*

*Proof.* For $b = 3$ the inequality is trivial: an admissible visit has at most $b$ vertices, so $M(\pi) \leq 3$ for every $\pi \in S_3$. For even $b \geq 6$, Corollary 5.5 gives $M(T_b) = 3$. For odd $b \geq 7$, let

$$T'_b \;=\; (0,\ 1,\ b{-}1,\ 3,\ 4,\ \ldots,\ b{-}2,\ 2) \in S_b$$

be the identity with the values 2 and $b{-}1$ exchanged – a sibling of $T_b$, which exchanges 1 and $b{-}1$. Appendix A classifies the admissible visits of $H_{T'_b}$ completely, in the manner of Theorem 5.4: up to reversal there are exactly five, of sizes $1, 3, 3, 3, 3$, so $M(T'_b) = 3$. (The parity mechanism is the same doubled-run obstruction as in Lemma 5.3, with the run shortened by one vertex, which is exactly what reverses the good parity from even $b$ to odd $b$.) The two remaining sizes $b = 4, 5$ are settled by the exhaustive enumeration already reported in Appendix D: $\min_{\pi \in S_b} M(\pi) = 3$ for $4 \leq b \leq 9$; for $b = 5$ the pattern $T'_5 = (0, 1, 4, 3, 2)$ attains it, with admissible-visit list $(0)$, $(0, 1, 2)$, $(0, 1, 4)$, $(2, 1, 4)$, $(2, 3, 4)$. □

## 7.2 The lower bound: every pattern admits a visit of size three

The lower bound cannot be proved by a local or greedy argument: for the identity permutation the *smallest* witness visit produced below is a terminal-to-terminal path through all $b$ vertices, so no argument that exhibits a bounded-size certificate can succeed uniformly. The proof instead builds the witness globally, from one matching inside each color class. Three elementary lemmas about matchings in paths do all the work. Throughout, $\pi \in S_b$ with $b \geq 3$; the *P-path* is the Hamiltonian path $0 - 1 - \cdots - (b-1)$ of P-edges and the *V-path* is the Hamiltonian path $\pi^{-1}(0) - \pi^{-1}(1) - \cdots - \pi^{-1}(b-1)$ of V-edges. For a matching $M$ contained in one of these paths, we say $M$ *exposes* the vertices it does not cover.

**Lemma 7.3** (Matching unions)**.** *Let $M_P$ be a matching contained in the P-edges and $M_V$ a matching contained in the V-edges, and let $U$ be their union, keeping colors (if a doubled pair contributes an edge to each, $U$ retains both parallel edges). Then:*

*(a) every vertex has degree $\leq 2$ in $U$; more precisely, the degree of $x$ in $U$ equals the number of the two matchings that cover $x$;*

*(b) the connected components of $U$ are simple paths (possibly single vertices) and cycles, and along any component the edges alternate between $M_P$ and $M_V$, hence alternate colors: every component is properly colored;*

*(c) the degree-1 vertices are exactly those covered by exactly one of the two matchings, and the unique $U$-edge at such a vertex belongs to the matching covering it. If $U$ has exactly $2\ell$ vertices of degree 1, then $U$ has exactly $\ell$ components that are paths with at least one edge, their ends pairing up the degree-1 vertices.*

*Proof.* (a) Each matching contributes at most one edge per vertex. Consecutive edges of a walk in $U$ share a vertex, so they cannot lie in the same matching; hence they alternate between $M_P$ and $M_V$, and since $M_P$-edges are P-colored and $M_V$-edges V-colored, the colors alternate, giving (b) (maximum degree 2 makes every component a path or a cycle; a doubled pair with both copies in $U$ is a 2-cycle component). The ends of the path components are the vertices of degree $\leq 1$, giving (c). □

**Lemma 7.4** (Prescribed exposure)**.** *Let $Q = x_0 - x_1 - \cdots - x_{m-1}$ be a path and let $S \subseteq \{x_0, \ldots, x_{m-1}\}$. A matching of $Q$ exposing exactly $S$ exists if and only if every maximal run of consecutive non-exposed vertices (in path order) has even size. In particular:*

*(a) $m$ odd: a matching exposing exactly $\{x_j\}$ exists iff $j$ is even;*

*(b) $m$ even: a perfect matching exists, and a matching exposing exactly $\{x_i, x_j\}$ ($i < j$) exists iff $i$ is even and $j$ is odd.*

*Proof.* If every run of non-exposed vertices is even, pair each run off consecutively. Conversely, a matching of a path covering a contiguous run of vertices bounded by exposed vertices must pair within the run, forcing even size. For (a), the runs have sizes $j$ and $m-1-j$, both even iff $j$ is even and $m$ is odd; for (b), the runs have sizes $i$, $j-i-1$, $m-1-j$, all even iff $i$ is even, $j$ is odd, $m$ is even. □

**Lemma 7.5** (Exposure trick)**.** *Suppose $M_P$ exposes exactly the vertex set $S_P$ within the P-path and $M_V$ exposes exactly $S_V$ within the V-path, and let $U$ be their union. Then:*

*(a) the degree-1 vertices of $U$ are exactly $(S_P \cup S_V) \setminus (S_P \cap S_V)$; the vertices of $S_P \cap S_V$ are isolated; all others have degree 2;*

*(b)* (opposite exposure $\Rightarrow$ odd size $\geq 3$) *if a path component of $U$ has one end $x \in S_P \setminus S_V$ and the other end $y \in S_V \setminus S_P$, then its end edges are V-colored at $x$ and P-colored at $y$, its edge count is even, and it has at least 2 edges: its size is odd and $\geq 3$;*

*(c)* (same exposure $\Rightarrow$ even size $\geq 2$) *if both ends lie in $S_P \setminus S_V$, the end edges are V-colored at both ends, the edge count is odd, and the size is even and $\geq 2$, with size exactly 2 precisely when the two ends are joined by a single $M_V$-edge (symmetrically with the colors exchanged for two ends in $S_V \setminus S_P$).*

*Proof.* (a) restates Lemma 7.3(a). For (b): $x \in S_P \setminus S_V$ is covered only by $M_V$, so its unique $U$-edge is V-colored; likewise the unique edge at $y$ is P-colored (Lemma 7.3(c)). Alternation with first color V and last color P forces an even number of edges, and the component has at least one edge since its ends have degree 1. If it had exactly one edge $e = \{x, y\}$, then $e$, being $x$'s edge, lies in $M_V$; but then $M_V$ covers $y$, contradicting $y \in S_V$. So the edge count is even and $\geq 2$, and the size (edges plus one) is odd and $\geq 3$. For (c): both end edges are V-colored, so alternation $\mathrm{V} \cdots \mathrm{V}$ gives an odd edge count $\geq 1$; the size is even and $\geq 2$, and size 2 is exactly the single-edge case, which forces that edge into $M_V$. ☐

**Theorem 7.6** (Floor lower bound)**.** *For every $\pi \in S_b$ with $b \geq 3$, the graph $H_\pi$ has a type-(i) admissible visit of size $\geq 3$. Hence $M(\pi) \geq 3$.*

*Proof.* Write $v_0 = \pi^{-1}(0)$ and $v_1 = \pi^{-1}(b-1)$; these are distinct vertices since $b \geq 2$. All matchings below live inside the P-path or the V-path, and their existence is by Lemma 7.4, using the following indices: vertex 0 has position-index 0 and vertex $b-1$ has position-index $b-1$ in the P-path; $v_0$ has value-index 0 and $v_1$ has value-index $b-1$ in the V-path.

**Case 1: $b$ odd.** Both P-path indices $0, b-1$ and both V-path indices $0, b-1$ are even. Set $c := v_1$ if $v_1 \neq 0$, and otherwise $c := v_0$; in the latter case $\pi(0) = b-1 \neq 0$, so $v_0 = \pi^{-1}(0) \neq 0$. Either way $c \neq 0$, and $c$ has V-path index 0 or $b-1$, both even. By Lemma 7.4(a) choose $M_P$ exposing exactly $\{0\}$ (concretely $\{1,2\}, \{3,4\}, \ldots, \{b-2, b-1\}$) and $M_V$ exposing exactly $\{c\}$. Then $S_P = \{0\}$ and $S_V = \{c\}$ are disjoint, so by Lemma 7.5(a) the union $U$ has exactly two degree-1 vertices, 0 and $c$, hence exactly one path component, from 0 to $c$. By Lemma 7.5(b) it is a properly colored path of odd size $\geq 3$ whose first edge is V-colored at vertex 0 and whose last edge is P-colored at $c$. Assign the role $p_0$ at vertex 0 (port color $\mathrm{P} \neq \mathrm{V}$) and the role $v_1$ or $v_0$ at $c$ (port color $\mathrm{V} \neq \mathrm{P}$): a type-(i) admissible visit of size $\geq 3$.

**Case 2: $b$ even and $\{\pi(0), \pi(b-1)\} \neq \{2k, 2k+1\}$ for every $k$.** By Lemma 7.4(b) choose $M_V$ to be the perfect matching of the V-path (value pairs $(0,1), (2,3), \ldots, (b-2, b-1)$) and $M_P$ exposing exactly $\{0, b-1\}$ (indices 0 even and $b-1$ odd). Then $S_P = \{0, b-1\}$, $S_V = \emptyset$: the union has exactly one path component, from 0 to $b-1$, with V-colored end edges at both ends, of even size $\geq 2$ (Lemma 7.5(c)). Size 2 would mean the single edge $\{0, b-1\} \in M_V$, i.e. $\{\pi(0), \pi(b-1)\}$ is one of the value pairs $\{2k, 2k+1\}$ – excluded by hypothesis. So the size is even and $\geq 4$; with roles $p_0, p_1$ (port colors P, end edges V) this is a type-(i) admissible visit of size $\geq 4$.

**Case 3: $b$ even and $\{\pi(0), \pi(b-1)\} = \{2k, 2k+1\}$ for some $k$.** By Lemma 7.4(b) choose $M_P$ exposing exactly $\{0, b-1\}$ and $M_V$ exposing exactly $\{v_0, v_1\}$ (value-indices 0 even and $b-1$ odd; the exposed value pairs are $(1,2), (3,4), \ldots, (b-3, b-2)$). Consider the coincidence set $T := \{0, b-1\} \cap \{v_0, v_1\}$.

*Case 3a: $T = \emptyset$.* There are four degree-1 vertices $0, b-1, v_0, v_1$, hence exactly two path components pairing them up. If some component joins a P-terminal to a V-terminal, Lemma 7.5(b) makes it an admissible visit of odd size $\geq 3$ (roles with opposite port colors at its ends) — done. Otherwise one component joins 0 to $b-1$ and the other joins $v_0$ to $v_1$, both of even size $\geq 2$ (Lemma 7.5(c)). If the $0 \rightsquigarrow b-1$ component had size 2, then $\{0, b-1\} \in M_V$, so $\{\pi(0), \pi(b-1)\}$ would be one of the value pairs $\{2j+1, 2j+2\}$ – an odd-aligned consecutive pair – while the case hypothesis makes it even-aligned; a two-element set of consecutive integers cannot be both. So that component has even size $\geq 4$, and with roles $p_0, p_1$ it is an admissible visit of size $\geq 4$ – done.

*Case 3b:* $|T| = 1$. Let $t$ be the coincident vertex ($t \in S_P \cap S_V$ is isolated in $U$), $x$ the other P-terminal and $y$ the other V-terminal; $x \neq y$ (else $|T| = 2$). The degree-1 vertices are exactly $x$ and $y$, so $U$ has one path component, from $x$ to $y$, which by Lemma 7.5(b) is an admissible visit of odd size $\geq 3$ (a P-role at $x$ with V-colored end edge, a V-role at $y$ with P-colored end edge) – done.

*Case 3c:* $|T| = 2$. Then $\{v_0, v_1\} = \{0, b-1\}$, i.e. $\{\pi(0), \pi(b-1)\} = \{0, b-1\}$, which is a set of consecutive integers only if $b = 2$ – contradicting $b \geq 4$ in the presence of the case-3 hypothesis. So this sub-case is vacuous.

Cases 1–3 cover all $\pi \in S_b$, $b \geq 3$. □

**Remark 7.7** (The witness is necessarily global)**.** For $\pi = \mathrm{id}_b$ with $b$ odd, Case 1 produces the full alternating path $0 - 1 - \cdots - (b-1)$: the witness has size $b$, not size 3. This is forced, not an artifact: in $H_{\mathrm{id}_b}$ every admissible visit joining terminals at the two ends must traverse the whole doubled path. A proof of Theorem 7.6 that exhibits a bounded-radius certificate around a terminal therefore cannot exist, which is why the matching-union construction, whose witness is a *component of a globally defined graph*, is the right tool.

*Proof of Theorem 7.1.* Combine Proposition 7.2 and Theorem 7.6; for $b = 2$, enumerate directly (each of the two patterns admits a size-2 visit between its two P-roles or V-roles, and size 3 is impossible on 2 vertices). □

**Remark 7.8** (Consequences for the method)**.** Theorem 7.1 answers the calibration question raised in Section 8: transit numbers cannot be pushed below 3, so within the inflation framework the per-visit cost of this paper's gadgets is optimal, and improvements to Theorem 1.1 must come from the visit *count* or from outside the inflation method.

**Remark 7.9** (The spectrum of transit numbers; data)**.** Exhaustive computation of $\{M(\pi) : \pi \in S_b\}$ for $b \leq 8$ (Appendix D) shows that the attained set is $\{3, 4, \ldots, b\}$ for even $b$ but $\{3\} \cup \{5, \ldots, b\}$ for odd $b$: the value 4 is not attained at odd $b \leq 8$. The maximum $M(\pi) = b$ is always attained (e.g. by the identity). We record the gap at 4 as data; we have no proof that it persists for odd $b > 8$.

# 8 Lower bounds and open problems

We work with $\pm 1$-blocks (Definition 2.3); $b(\sigma)$ is their number and $s_{\max}(\sigma)$ the largest size. What is known on the lower-bound side is quickly stated. The only unconditional lower bound we have is the floor $\rho \geq s_{\max}$ (Proposition 2.4), and it degenerates exactly where it is needed most: on permutations whose blocks all have bounded size, $s_{\max}$ is constant while $n$ grows. Beyond the floor our methods produce no lower bounds at all, while Theorem 1.1 caps everything at $O(\sqrt{n}\,)$; we expect the cap to be the truth, i.e. that $\rho_{\min}(n) = \Theta(\sqrt{n})$. This is the state of the problem we most want to advertise:

*Main open problem. Prove or disprove:* $\rho_{\min}(n) \to \infty$.

The difficulty is easy to underestimate. The graphs $H_\sigma$ are spanning, connected, of maximum degree four, and every vertex lies on two Hamiltonian monochromatic paths; the natural intuition is that so much connectivity must force alternating paths of unbounded length. We know no quantity that makes that intuition precise. A disproof – a family with $\rho$ bounded – would be at least as interesting, though we do not expect one.

On the gadget side, the calibration question is settled by Theorem 7.1: transit numbers cannot be made smaller than 3 at any size, so inflation with better blocks cannot improve the method's constants, and (heuristically) inflation alone cannot beat $\sqrt{n}$: the end-visit term ties the bound to the block size, and nesting inflations doubles the visit-count bound per level. What remains open there is the fine structure of the transit spectrum (Remark 7.9): is the value 4 unattained at every odd $b$?

Three further directions, each supported by exact data.

**More colors.** For $k$ permutations $w_1, \ldots, w_k$ of a common vertex set, color the edges of the $k$ Hamiltonian paths $P(w_1), \ldots, P(w_k)$ with $k$ colors, one per path, and let $\rho^{(k)}_{\min}(n)$ denote the minimum over all $k$-tuples of the maximum number of vertices on a properly colored simple path. (For $k = 2$ this is $\rho_{\min}(n)$: anchoring one path as the identity is a relabeling.) Adding a color only adds edges, so

$$\rho_{\min}(n) \;=\; \rho^{(2)}_{\min}(n) \;\leq\; \rho^{(3)}_{\min}(n) \;\leq\; \cdots \;\leq\; n.$$

Exhaustive computation (ancillary artifact) gives $\rho^{(3)}_{\min}(n) = n$ for $4 \leq n \leq 10$: *every* union of three Hamiltonian paths on at most ten vertices has a properly colored Hamiltonian path – in particular a third ordering erases the two-color dips at $n = 7, 8$. But three orderings do not suffice forever:

**Proposition 8.1.** $\rho^{(3)}_{\min}(11) = 10$. *Up to symmetry (permuting the three paths, reversing paths, and relabeling) there is exactly* one *triple of Hamiltonian paths on* 11 *vertices whose union has no properly colored Hamiltonian path, namely* $w_1 = \mathrm{id}$, $w_2 = (0, 1, 2, 3, 4, 6, 7, 8, 5, 9, 10)$, $w_3 = (0, 1, 5, 2, 3, 4, 6, 7, 8, 9, 10)$ *(the identity together with one-step cyclic shifts of the vertex intervals* $[5, 8]$ *and* $[2, 5]$*); this triple attains* $\rho = 10$*. The count over anchored unordered pairs* $\{w_2, w_3\}$ *is* 12*, all in one symmetry orbit.*

The computation is a census, not a triple-by-triple sweep: a properly colored path of any two of the three colors survives in the union, so a deficient triple $(\mathrm{id}, s, t)$ requires all three of its two-color *shadows* – the pairs $(\mathrm{id}, s)$, $(\mathrm{id}, t)$, and $(s, t) \cong (\mathrm{id}, s^{-1}t)$ – to be deficient; it therefore suffices to enumerate coset triangles $\{s,\, t,\, s^{-1}t\}$ inside the deficient set $D(n) = \{w : \rho(\mathrm{id}, w) \leq n - 1\}$ and to decide the surviving candidates directly. All twelve witnessing pairs were certified independently by an integer-programming formulation. We find the contrast with $k = 2$ striking: at $n = 11$ there are 135,712 deficient permutations and $3.9 \times 10^8$ candidate triples, yet a single triple (up to symmetry) survives, and its deficit is exactly one vertex.

> *Open problem (the $k$-color hierarchy). Does $\rho^{(3)}_{\min}(n) \to \infty$? Is $\rho^{(3)}_{\min}(n) = n - O(1)$? More generally, does $\rho^{(k)}_{\min}(n) = O(\sqrt{n})$ – or even $o(n)$ – hold for every fixed $k$, as it does for $k = 2$, or is there a $k$ for which $\rho^{(k)}_{\min}(n) \geq cn$?*

Nothing we know separates these possibilities: the gadget calculus of Sections 3–5 is genuinely two-colored (an interval block of one permutation need not be an interval block of another), and no analogue of the transit obstruction is currently available for $k \geq 3$.

**The matching-edge parameter.** Second, the matching-edge parameter of the scrambled ladder (Appendix C): how does $f(n) = \min_\sigma \max_\pi \mu(\pi)$ grow? Exhaustive computation of $\max_\pi \mu(\pi)$ over all of $S_n$ for $n \le 9$ (ancillary artifact) gives the first exact values:

$$f(n) = n \quad (2 \le n \le 6), \qquad f(7) = 6, \quad f(8) = 7, \quad f(9) = 8,$$
$$f(10) = 8, \quad f(11) = 9, \quad f(12) = 10,$$

and, in fact, $\max_\pi \mu(\pi) \in \{n-1, n\}$ for *every* $\sigma \in S_n$, $n \le 9$ – but not beyond: from $n = 10$ a third value appears. Exactly 8, 24, and 1088 permutations of $S_{10}$, $S_{11}$, $S_{12}$ (respectively) have $\max_\pi \mu = n - 2$, and none does worse through $n = 12$, so $f(n) = n - 2$ there. The smallest examples are the four-block permutations of shape $(2, 3, 3, 2)$, e.g. $\sigma = (1, 0,\ 4, 3, 2,\ 7, 6, 5,\ 9, 8)$, where two odd blocks each cost a rung; all exceptional permutations satisfy $\rho(\sigma) \le n - 2$, as the conversion inequality forces. The smallest $n$ admitting a three-rung loss lies between 13 and 29, the upper end from the exact integer-programming values for the chain and gadget families. The top value is exact and structural:

**Observation 8.2.** $\max_\pi \mu(\pi) = n$ if and only if $\rho(\sigma) = n$. Indeed, a simple path in $G(n, \sigma)$ using all $n$ matching edges visits all $2n$ vertices and has exactly $n - 1$ non-matching edges, one between matching edges that are consecutive along the path; consecutive connector edges must alternate between the top and bottom paths, so the path is precisely the image of a properly colored Hamiltonian path under the conversion of Theorem C.2, and pulls back. Thus the conversion inequality $\max_\pi \mu \ge \rho$ is tight at full content, and $f(n) \le n - 1$ for every $n \ge 7$.

On the negative side, the collapse of $\rho$ does *not* transfer: the gadget family retains high matching content at every point we can compute exactly. Exact values (integer programming, optimality certified) are

$$\max_\pi \mu \;=\; n,\ n-1,\ n-2,\ n-3,\ n-4 \qquad \text{for } \sigma_{a,6},\ a = 2, \ldots, 6,$$

i.e. the loss is $a - 2$ – one matching edge per gadget copy beyond the second – while $\rho(\sigma_{a,b}) = \Theta(a + b)$ collapses; the snake bound $\ge n - 3a$ of `verify_f_linear.py` is far from optimal.

**Conjecture 8.3.** $f(n) = \Theta(n)$*: there is a constant* $c > 0$ *with* $f(n) \ge cn$ *for all* $n$*.*

The strongest lower bound known is $f(n) \ge 0.1\, n^{4/5}$ (Appendix C), so even linearity is open. Two more ambitious targets suggest themselves: $f(n) = n - o(n)$ (sublinear loss), and the sharper $f(n) \ge n - O(\sqrt{n})$, which would make the loss of $f$ match the collapse scale of $\rho$ itself. The exact values above mark the fork: if the one-edge-per-copy pattern of $\sigma_{a,6}$ persists for every $a$, then $f(n) \le \frac{5}{6}\, n + 2$ infinitely often, refuting both targets while leaving Conjecture 8.3 intact; if it saturates, the ambitious targets come back into play. Settling the pattern for large $a$ – a single structured family – may be more tractable than the conjecture itself.

**The exact gadget value.** Third, the machine computations of Appendix D support promoting the observed formula for the gadget family to a conjecture:

**Conjecture 8.4.** $\rho(\sigma_{a,b}) = 2a + 2b - 4$ *for all* $a \ge 3$ *and even* $b \ge 6$*.*

This holds at every tested point up to $n = 720$; the two-sided estimate (1) brackets it. Closing the gap between $2a + 2b - 7$ and $2a + 2b - 4$ is a self-contained exercise in the port calculus and would make a clean first project on these objects.

## A The odd gadget: classification of admissible visits of $H_{T'_b}$

This appendix proves the odd-size half of Proposition 7.2: for every odd $b \geq 7$, the permutation

$$T'_b \;=\; (0,\ 1,\ b{-}1,\ 3,\ 4,\ \ldots,\ b{-}2,\ 2) \in S_b \qquad \big(T'_b(2) = b{-}1,\ T'_b(b{-}1) = 2,\ T'_b(i) = i \text{ otherwise}\big)$$

has $M(T'_b) = 3$, by a complete classification of its admissible visits in the manner of Theorem 5.4. $T'_b$ is an involution; $(T'_b)^{-1}$ fixes 0, 1 and each $i \in [3, b-2]$ and swaps $2 \leftrightarrow b-1$. The terminals (Definition 3.3) are

$$p_0 = v_0 = \text{vertex } 0, \qquad v_1 = (T'_b)^{-1}(b-1) = \text{vertex } 2, \qquad p_1 = \text{vertex } b-1.$$

Note the contrast with $T_b$: there the swap $1 \leftrightarrow b-1$ makes vertex 1 the terminal $v_1$; here the swap $2 \leftrightarrow b-1$ moves $v_1$ to vertex 2, one step further from the end of the position path, and this relabeling is the entire source of the parity reversal that makes the construction work at odd $b$.

**Lemma A.1** (Inventory of $H_{T'_b}$)**.** *For odd $b \geq 7$, the edges of $H_{T'_b}$ are exactly:*

- *doubled pairs* $\{0,1\}$ *and* $\{k, k+1\}$ *for* $3 \leq k \leq b-3$*;*
- *P-only edges* $\{1,2\}$*,* $\{2,3\}$*,* $\{b-2, b-1\}$*;*
- *V-only edges* $\{1, b-1\}$*,* $\{2, b-2\}$*,* $\{3, b-1\}$*.*

*The full edge lists at the special vertices are:*

| | |
|---|---|
| *vertex* $0$ : | *doubled* $\{0,1\}$ *only (a leaf: its unique neighbor is vertex* $1$*),* |
| *vertex* $1$ : | $\mathrm{P}\,\{0,1\},\ \mathrm{P}\,\{1,2\};\ \ \mathrm{V}\,\{0,1\},\ \mathrm{V}\,\{1,b{-}1\}$, |
| *vertex* $2$ : | $\mathrm{P}\,\{1,2\},\ \mathrm{P}\,\{2,3\};\ \ \mathrm{V}\,\{2,b{-}2\}$ (*unique V-edge*), |
| *vertex* $3$ : | $\mathrm{P}\,\{2,3\}$, *doubled* $\{3,4\}$; $\mathrm{V}\,\{3,b{-}1\}$, |
| *vertex* $b{-}2$ : | *doubled* $\{b{-}3,b{-}2\}$; $\mathrm{P}\,\{b{-}2,b{-}1\};\ \ \mathrm{V}\,\{2,b{-}2\}$, |
| *vertex* $b{-}1$ : | $\mathrm{P}\,\{b{-}2,b{-}1\};\ \ \mathrm{V}\,\{1,b{-}1\},\ \mathrm{V}\,\{3,b{-}1\}$ (*unique P-edge*), |
| *vertex* $i,\ 4 \leq i \leq b-3$ : | *doubled* $\{i{-}1,i\}$ *and* $\{i,i{+}1\}$ *only.* |

*Proof.* P-edges are $\{i, i+1\}$, $0 \leq i \leq b-2$. The V-edges, indexed by $v = 0, \ldots, b-2$, are $\{(T'_b)^{-1}(v), (T'_b)^{-1}(v+1)\}$:

$$v = 0:\ \{0,1\}; \qquad v = 1:\ \{1, b{-}1\}; \qquad v = 2:\ \{b{-}1, 3\};$$
$$3 \leq v \leq b-3:\ \{v, v{+}1\}; \qquad v = b-2:\ \{b{-}2, 2\}.$$

The consecutive-position pairs appearing in both lists are exactly $\{0,1\}$ and $\{k, k+1\}$ for $3 \leq k \leq b-3$: doubled. The remaining V-edges $\{1, b-1\}, \{3, b-1\}, \{2, b-2\}$ join non-adjacent positions for $b \geq 7$: V-only. The remaining P-edges $\{1,2\}, \{2,3\}, \{b-2, b-1\}$ are P-only. The per-vertex lists follow by inspection. $\square$

We use three forcing facts, the analogues of (F1)–(F3) in the proof of Theorem 5.4: **(F0)** vertex 0 is a leaf whose unique neighbor is vertex 1, reached by the doubled pair $\{0,1\}$ – so in any simple PC path, vertex 0 is an endpoint; **(F1)** vertex $b-1$ has a unique P-edge, $\{b-2, b-1\}$; **(F2)** vertex 2 has a unique V-edge, $\{2, b-2\}$.

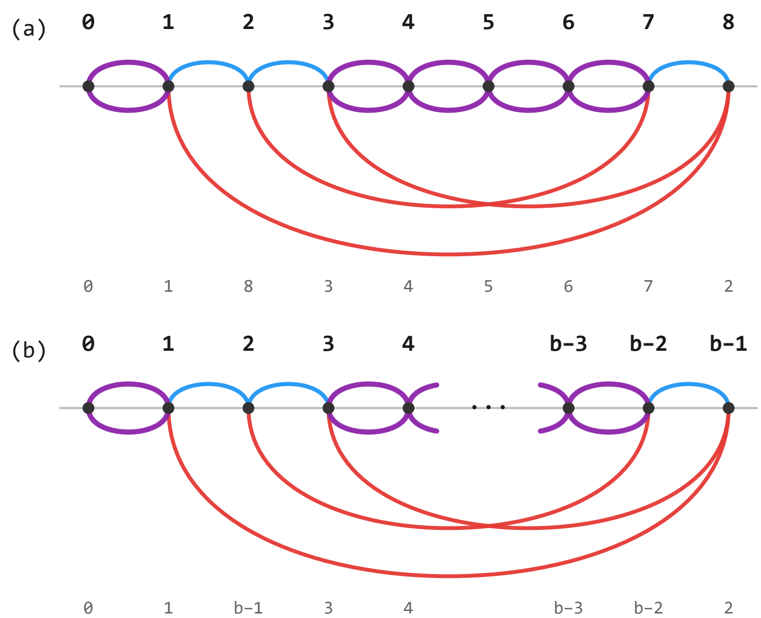


Figure 2: The odd gadget $H_{T_b'}$: (a) the instance $T_9' = (0, 1, 8, 3, 4, 5, 6, 7, 2)$; (b) general odd $b$, the middle of the highway elided. Conventions as in Figure 1. Note the contrasts with $H_{T_b}$: vertex 0 is a leaf (its doubled pair $\{0, 1\}$ is its only edge pair), the terminal $v_1$ sits at vertex 2, and the doubled run $D = \{3, \ldots, b-2\}$ is one vertex shorter than the segment of $H_{T_b}$ – the relabeling that reverses the good parity from even to odd $b$.

Let $D = \{3, 4, \ldots, b-2\}$ (the *highway*), $|D| = b-4$. By Lemma A.1, the sub-multigraph on $D$ is the doubled path $3 - 4 - \cdots - (b-2)$; the only edges from $D$ to its complement are the two at vertex 3 (P $\{2, 3\}$, V $\{3, b-1\}$) and the two at vertex $b-2$ (P $\{b-2, b-1\}$, V $\{2, b-2\}$); and no terminal lies in $D$. Figure 2 pictures the inventory.

**Lemma A.2** (Highway behavior)**.** *Let $Q$ be a PC path in $H_{T_b'}$, $b \geq 7$ odd.*

*(a) While inside $D$, $Q$ moves monotonically in position, and $Q$ can leave $D$ only from vertex 3 or vertex $b-2$.*

*(b) Suppose $Q$ enters $D$ at one end (3 or $b-2$) by an edge of color $c$, at a moment when no vertex of $D$ has yet been visited, and later leaves $D$. Then either it leaves immediately from the entry vertex by the other external edge there, of color $\bar{c}$, or it traverses all of $D$ to the far end; in the latter case its $b-5$ internal edges alternate colors starting with $\bar{c}$, and since $b$ is* odd, *$b-5$ is* even, *so the last internal edge has color $c$ and any edge by which $Q$ then leaves $D$ has color $\bar{c}$.*

*Proof.* Identical in form to Lemma 5.3, applied to $D$: (a) is immediate since a simple path cannot reverse inside a doubled path graph and the boundary vertices of $D$ are 3 and $b-2$; the two external edges at each of 3 and $b-2$ have different colors, so an immediate exit uses the other one, of color $\bar{c}$. A full crossing has $b-5$ internal edges (a path on $b-4$ vertices); the PC condition at the entry gives the first internal edge color $\bar{c}$, alternation gives the last internal edge color $c$ (as $b-5$ is even), and the PC condition at the far end forces any exit edge to color $\bar{c}$. $\square$

Note the parity contrast with $T_b$: the segment of Lemma 5.3 has $b-4$ internal edges (even iff $b$ even), while the highway here has $b-5$ (even iff $b$ odd), because the relabeling detached vertex 2 from the doubled run. The parity that serves even $b$ in Theorem 5.4 is exactly the parity that serves odd $b$ here.

**Theorem A.3** (Classification, odd gadget)**.** *Let $b \geq 7$ be odd. Up to reversal, the admissible visits of $H_{T'_b}$ are exactly the following, each with the role pair shown:*

| | *vertices* | *edges (colors)* | *role pair* |
|---|---|---|---|
| *(W1)* | 0 | — | $\{p_0, v_0\}$ |
| *(W2)* | 0, 1, 2 | V$\{0,1\}$, P$\{1,2\}$ | $\{p_0, v_1\}$ |
| *(W3)* | 0, 1, $b-1$ | P$\{0,1\}$, V$\{1,b{-}1\}$ | $\{v_0, p_1\}$ |
| *(W4)* | 2, 1, $b-1$ | P$\{1,2\}$, V$\{1,b{-}1\}$ | $\{v_1, p_1\}$ |
| *(W5)* | 2, 3, $b-1$ | P$\{2,3\}$, V$\{3,b{-}1\}$ | $\{v_1, p_1\}$ |

*In particular every admissible visit has at most 3 vertices, and $M(T'_b) = 3$, attained by (W2)–(W5).*

*Proof. Each listed visit is admissible.* (W1): vertex 0 carries $p_0$ and $v_0$ with different port colors – type (ii). (W2): the visit $(0,1,2)$ takes the V-copy of the doubled pair $\{0,1\}$, then P$\{1,2\}$; first edge V $\neq \mathrm{pc}(p_0) =$ P, last edge P $\neq \mathrm{pc}(v_1) =$ V. (W3): P-copy of $\{0,1\}$, then V$\{1,b-1\}$; first P $\neq \mathrm{pc}(v_0)$, last V $\neq \mathrm{pc}(p_1)$. (W4): P$\{1,2\}$ then V$\{1,b-1\}$; first P $\neq \mathrm{pc}(v_1)$, last V $\neq \mathrm{pc}(p_1)$. (W5): P$\{2,3\}$ then V$\{3,b-1\}$; the same check. The role pairs are forced by the first/last edge colors as in Theorem 5.4. (Note vertex 1 is *not* a terminal here – it is an interior vertex of (W2)–(W4).)

*No others exist.* A type-(ii) visit needs a vertex carrying two roles of different port colors; the only vertex carrying two roles is $0 = p_0 = v_0$: (W1). A type-(i) visit starts at a terminal vertex with a constrained first color (Lemma 3.5(b)); the terminal vertices are $0, 2, b-1$, and the possible (start, first color) stems are

$$\text{(A) } (0, \text{V})\ [p_0], \quad \text{(B) } (0, \text{P})\ [v_0], \quad \text{(C) } (2, \text{P})\ [v_1], \quad \text{(D) } (b-1, \text{V})\ [p_1]$$

(vertex 2 carries only $v_1$, which requires first color P; vertex $b-1$ carries only $p_1$, requiring first color V). We enumerate the PC continuations of each stem, recording every moment the path stands at a terminal vertex with an admissible last-edge color; since no vertex of $D$ is a terminal, a path stranded inside $D$ ends no admissible visit.

**Stem (A): start** 0**, first edge** V. By (F0) the first edge is the V-copy of $\{0,1\}$; at 1, last color V; no role at 1. Continue with a P-edge: $\{0,1\}$ is excluded (0 visited), so P$\{1,2\}$; at 2, last color P. *Record:* $v_1$ at 2 has pc = V $\neq$ P: this is (W2). Continue: next $\neq$ P, so by (F2) V$\{2, b-2\}$, entering $D$ at $b-2$ with color V, $D$ untouched. By Lemma A.2(b): *immediate exit* by P$\{b-2, b-1\}$: at $b-1$, last color P; the only role there is $p_1$ with pc = P: not admissible. Continue: next $\neq$ P, a V-edge at $b-1$: $\{1, b-1\}$ has 1 visited, so V$\{3, b-1\}$ into $D$ at 3; from 3 the non-V continuations are P$\{2,3\}$ (2 visited) or the P-side of $\{3,4\}$, after which the path ascends monotonically and strands ($b-2$ visited). *Full crossing* to 3: arrival color V, so the exit must be P; the only external P-edge at 3 is $\{2,3\}$, with 2 visited. Strands. *Stem (A) yields exactly (W2).*

**Stem (B): start** $0$**, first edge** P**.** By (F0) the first edge is the P-copy of $\{0,1\}$; at 1, last P; no role. Continue with a V-edge: $\{0,1\}$ excluded, so V$\{1,b-1\}$; at $b-1$, last V. *Record:* $p_1$ admissible: (W3). Continue: next $\neq$ V, by (F1) P$\{b-2,b-1\}$ into $D$ at $b-2$, color P, $D$ untouched. *Immediate exit* by V$\{2,b-2\}$: at 2, last V; role $v_1$ has pc $=$ V: not admissible. Continue: next $\neq$ V, a P-edge $\{1,2\}$ (1 visited) or P$\{2,3\}$ into $D$ at 3; from 3, next $\neq$ P is V$\{3,b-1\}$ ($b-1$ visited) or the V-side of $\{3,4\}$, then monotone ascent strands ($b-2$ visited). *Full crossing* to 3: arrival P, exit must be V; only $\{3,b-1\}$, with $b-1$ visited. Strands. *Stem (B) yields exactly (W3).*

**Stem (C): start** $2$**, first edge** P**.** The P-edges at 2 are $\{1,2\}$ and $\{2,3\}$. *Branch* $2\to 1$*:* at 1, last P; no role. Continue with a V-edge: either V$\{0,1\}$ – at 0, last V; *record:* $p_0$ (pc $=$ P) admissible: (W2) reversed; by (F0) the only continuation returns to 1, visited: dead end – or V$\{1,b-1\}$ – at $b-1$, last V; *record:* $p_1$ admissible: (W4). Continue: next $\neq$ V, by (F1) P$\{b-2,b-1\}$ into $D$ at $b-2$, color P, $D$ untouched; immediate exit by V$\{2,b-2\}$ has 2 visited; full crossing to 3 arrives with P, exit must be V, only $\{3,b-1\}$ with $b-1$ visited. Strands. *Branch* $2\to 3$*:* enters $D$ at 3, color P, $D$ untouched. Immediate exit by V$\{3,b-1\}$: at $b-1$, last V; *record:* $p_1$ admissible: (W5). Continue: next $\neq$ V, by (F1) P$\{b-2,b-1\}$ into $D$ at $b-2$ – but $D$ is now touched (vertex 3): the non-P continuations at $b-2$ are V$\{2,b-2\}$ (2 visited) and the V-side of $\{b-3,b-2\}$, then monotone descent strands at 4 (3 visited). Alternatively, full crossing $3\to b-2$ arrives with color P; the exit must be V, and the only external V-edge at $b-2$ is $\{2,b-2\}$, with 2 visited. Strands. *Stem (C) yields exactly (W2) reversed, (W4), (W5).*

**Stem (D): start** $b-1$**, first edge** V**.** The V-edges at $b-1$ are $\{1,b-1\}$ and $\{3,b-1\}$. *Branch* $b-1\to 1$*:* at 1, last V; no role. Continue with a P-edge: either P$\{0,1\}$ – at 0, last P; *record:* $v_0$ (pc $=$ V) admissible: (W3) reversed; the only continuation returns to 1: dead end – or P$\{1,2\}$ – at 2, last P; *record:* $v_1$ admissible: (W4) reversed. Continue: next $\neq$ P, by (F2) V$\{2,b-2\}$ into $D$ at $b-2$, color V, $D$ untouched; immediate exit by P$\{b-2,b-1\}$ has $b-1$ visited; full crossing to 3 arrives with V, exit must be P, only $\{2,3\}$ with 2 visited. Strands. *Branch* $b-1\to 3$*:* enters $D$ at 3, color V, $D$ untouched. Immediate exit by P$\{2,3\}$: at 2, last P; *record:* $v_1$ admissible: (W5) reversed. Continue: next $\neq$ P, by (F2) V$\{2,b-2\}$ into $D$ at $b-2$ – $D$ touched (vertex 3): the non-V continuations at $b-2$ are P$\{b-2,b-1\}$ ($b-1$ visited) and the P-side of $\{b-3,b-2\}$, then monotone descent strands at 4. Alternatively, full crossing $3\to b-2$ arrives with color V; the exit must be P, only $\{b-2,b-1\}$, with $b-1$ visited. Strands. *Stem (D) yields exactly (W3) reversed, (W4) reversed, (W5) reversed.*

Collecting the stems: every admissible visit is one of (W1)–(W5) up to reversal, and the maximum size is 3. □

**Remark A.4** (Degenerate and small cases)**.** For $b=5$ the highway $D=\{3\}$ degenerates and Lemmas A.1–A.2 do not apply verbatim, but direct enumeration confirms that the classification's *shape* persists: $T_5'=(0,1,4,3,2)$ has admissible visits exactly $(0)$, $(0,1,2)$, $(0,1,4)$, $(2,1,4)$, $(2,3,4)$, so $M(T_5')=3$. For $b=3,4$ no member of either gadget family exists, and the exhaustive minima $\mu(3)=\mu(4)=3$ are computed directly (Appendix D). Note also the role pairs realized here — $\{p_0,v_1\}$ and $\{v_0,p_1\}$ appear, while $T_b$ forbids exactly those two — so the two gadget families attain the transit floor through genuinely different visit structures.

## B A $\rho_{\min}$ census

For $n \le 12$ the exact value of $\rho_{\min}(n)$, and the size of the *deficient set* $D(n) = \{\sigma \in S_n : \rho(\sigma) < n\}$, are known by exhaustive enumeration (the joint $(b, \rho)$ censuses of Appendix D, logs `exh7.txt`–`exh12.txt`). For $13 \le n \le 16$ the value of $\rho_{\min}(n)$ is likewise exact, by exhaustive *decision censuses at a threshold*: the search enumerates one representative per orbit of the $\rho$-preserving symmetries of $H_\sigma$ (reversal of $\sigma$, value-complementation, and inversion, the last of which exchanges the two colors) and records every $\sigma$ at or below the threshold. This determines $\rho_{\min}(n)$ and *all* of its minimizers, but not the full distribution, so $|D(n)|$ is not recorded in that range. Every minimizer found was certified by the same independent integer-programming formulation used for Proposition 8.1; these censuses and their certificates are archived in the research repository rather than in the ancillary artifact. For $n \le 6$ every union is properly-colored Hamiltonian: $D(n) = \emptyset$ and $\rho_{\min}(n) = n$, which also follows from $f(n) = n$ for $n \le 6$ together with Observation 8.2.

| $n$ | $n!$ | $\lvert D(n)\rvert$ | $\rho_{\min}(n)$ |
|---|---|---|---|
| $\le 6$ | | 0 | $n$ |
| 7 | 5,040 | 8 | 5 |
| 8 | 40,320 | 32 | 6 |
| 9 | 362,880 | 896 | 7 |
| 10 | 3,628,800 | 5,824 | 8 |
| 11 | 39,916,800 | 135,712 | 7 |
| 12 | 479,001,600 | 1,178,624 | 8 |
| 13 | 6,227,020,800 | — | 9 |
| 14 | 87,178,291,200 | — | 10 |
| 15 | 1,307,674,368,000 | — | 9 |
| 16 | 20,922,789,888,000 | — | 10 |

Three features of the table are worth recording. First, $\rho_{\min}$ is not monotone: the values at $n = 7, 11, 15$ sit below both neighbors. Second, the deficient sets remain a vanishing fraction of $S_n$ throughout the range where the full distribution is known ($|D(12)|/12! \approx 0.25\%$), while their internal structure is what the three-color census of Proposition 8.1 mines; the minimizers themselves are rare (the number of $\sigma$ attaining $\rho_{\min}(n)$ at $n = 13, 14, 15, 16$ is 290, 2894, 8, 240 respectively). Third, the deficient set is exactly where the ladder parameter of Appendix C dips: $\max_\pi \mu(\pi) \le n-1$ if and only if $\sigma \in D(n)$ (Observation 8.2), and for $n \le 12$ the dip is one rung except on the small exceptional sets of Section 8 (8, 24, 1088 permutations at $n = 10, 11, 12$), where it is exactly two. These are data, not theorems; nothing in the main text depends on this table beyond the values $\rho_{\min}(7) = 5$ and $\rho_{\min}(8) = 6$ and the counts $|D(n)|$ quoted in Section 8.

## C The scrambled ladder and the parameter $f(n)$

This appendix defines the graph family $G(n, \sigma)$ – the *scrambled ladder* – and its matching-edge parameter $f(n)$, and proves the conversion lemma linking $f$ to $\rho_{\min}$. It is logically independent of the main text; the growth of $f$ is posed among the open problems of Section 8.

**Definition C.1.** For $\sigma \in S_n$, the *scrambled ladder* $G(n, \sigma)$ is the graph on the $2n$ vertices $v_0, \ldots, v_{n-1}$, $u_0, \ldots, u_{n-1}$, with edges: a *top path* $v_0 v_1 \cdots v_{n-1}$; a *bottom path* $u_0 u_1 \cdots u_{n-1}$; and a *matching* $M = \{v_i u_{\sigma(i)} : 0 \le i \le n-1\}$. Every $v_i$ and every $u_j$ is incident to exactly one matching edge; the four path endpoints have degree 2 and every other vertex degree 3. For a simple path $\pi$ in $G(n,\sigma)$, $\mu(\pi)$ is the number of edges of $\pi$ in $M$, and $f(n) = \min_\sigma \max_\pi \mu(\pi)$.

**Theorem C.2** (Conversion Lemma)**.** *If $H_\sigma$ has a PC path on $k$ vertices, then $G(n,\sigma)$ has a simple path on $2k$ vertices using exactly $k$ matching edges. Hence* $\max_\pi \mu(\pi) \ge \rho(\sigma)$ *for every $\sigma$, and $f(n) \ge \rho_{\min}(n)$.*

*Proof.* Let $w_0 w_1 \cdots w_{k-1}$ be a PC path in $H_\sigma$ with edge colors $c_0, \ldots, c_{k-2}$ alternating. Build a path $\Pi$ in $G(n,\sigma)$ on the $2k$ vertices $\{v_{w_i}\} \cup \{u_{\sigma(w_i)}\}$ as follows. If $c_0 = \mathrm{P}$, run

$$u_{\sigma(w_0)} \xrightarrow{M} v_{w_0} \xrightarrow{P} v_{w_1} \xrightarrow{M} u_{\sigma(w_1)} \xrightarrow{Q} u_{\sigma(w_2)} \xrightarrow{M} v_{w_2} \xrightarrow{P} v_{w_3} \xrightarrow{M} \cdots$$

alternating matching edges with top/bottom path edges; if $c_0 = \mathrm{V}$, start instead at $v_{w_0}$, symmetrically.

*Edge validity.* A P-edge $\{w_i, w_{i+1}\}$ means $|w_i - w_{i+1}| = 1$, so $v_{w_i} v_{w_{i+1}}$ is a top-path edge; a V-edge $\{w_i, w_{i+1}\}$ means $|\sigma(w_i) - \sigma(w_{i+1})| = 1$, so $u_{\sigma(w_i)} u_{\sigma(w_{i+1})}$ is a bottom-path edge. The color alternation of the PC path matches the alternation of top versus bottom traversal in $\Pi$, so every step of $\Pi$ is a genuine edge.

*Simplicity.* The top vertices $\{v_{w_i}\}$ are distinct because the $w_i$ are; the bottom vertices $\{u_{\sigma(w_i)}\}$ are distinct because $\sigma$ is a bijection; the two families are disjoint.

*Matching count.* $\Pi$ uses one matching edge per vertex of the PC path, namely $v_{w_i} u_{\sigma(w_i)}$ for $0 \le i < k$: a total of $k$.

Taking $\pi = \Pi$ gives $\max_\pi \mu(\pi) \ge \rho(\sigma)$ for each $\sigma$; the minimum over $\sigma$ gives $f(n) \ge \rho_{\min}(n)$. $\square$

The conversion is one-directional: a path in $G(n,\sigma)$ that uses two consecutive edges of the same path (top or bottom) has no PC-path preimage, so the inequality $f(n) \ge \rho_{\min}(n)$ cannot be reversed. This is the precise sense in which the negative result for $\rho_{\min}$ (Theorem 1.1) does not transfer to $f$.

What is known about $f$ beyond the conversion inequality is quickly stated. The strongest lower bound we are aware of is $f(n) \ge 0.1\, n^{4/5}$, which is Lemma 8 of [4] (there stated for an arbitrary matching between two vertex-disjoint paths; suppressing unmatched vertices makes the two formulations equivalent), resting on the circumference theorem of [5]; Observation 8.2 gives $f(n) \le n - 1$ for $n \ge 7$, and the first exact values appear in Section 8. Historically, this construction is how the problem of this paper arose: it relates to the Hamiltonicity question of Lovász [7] for vertex-transitive graphs through Lemma 8 of [4]; although that route is now closed – sublinear-expander methods reach the same $n^{2/3-o(1)}$ target without the matching structure [6] – it was the initial motivation for our reduction into the study of $\rho_{\min}(n)$.

# D Computational verification

The asymptotic theorems and their constants are proved without computation; the computer-assisted claims are the small-size base cases $b \in \{4, 5\}$ of Proposition 7.2, the spectrum data of Remark 7.9, the $\rho_{\min}$ census of Appendix B, and the $k$-color and matching-content values of Section 8. This section specifies the computations that independently verify the constructions and supply the exact small-case values, and records the empirical sharpness of the families. **All code, command lines, and output logs are provided as ancillary files with this**

**submission** (directory `code/` in the repository; `code/README.md` maps each claim below to a script and a log, and `code/MANIFEST.sha256` lists SHA-256 checksums).

**Algorithm.** The basic object is an exact longest-PC-path search. Its state space is the set of triples (*visited set*, *current vertex*, *last edge color*); the initial states are all $(\{x, w\}, w, c)$ over edges $\{x, w\}$ of color $c$; a transition extends a state by an unvisited neighbor along an edge of the other color. The legal future extensions of a partial PC path depend only on its state, so distinct PC paths reaching the same state may be identified without loss; $\rho(\sigma)$ is then the maximum cardinality of the visited set over all *reachable* states, computed by an exhaustive search that visits every reachable state once (a memo set of states; no pruning). The method is exact by construction; its cost is the number of reachable states. That number is exponential in general – on unstructured instances we used the method only for $n \leq 19$ – but is small on the structured families here, because the gadget admits relatively few reachable states: for $\sigma_{a,b}$ we measure 57,066 reachable states at $n = 80$, 5,012,186 at $n = 600$, and 7,273,586 at $n = 720$ (the logs report the count for every table row). The constrained variants computing $M(\pi)$ (fixed start/end terminals and first/last colors) are the same search with the initial and accepting states restricted.

**Implementations.** Three independent implementations were used: a plain recursive enumerator and a memoized iterative one (both Python, in the artifact), which agree on 300 random permutations of sizes 4–8; and an independently written C++ solver (also in the artifact), which agrees with the Python solvers on every instance checked by both, including exact values of $\rho(\sigma_{a,b})$ at $(a, b) \in \{(6, 8), (5, 10), (4, 12)\}$. In addition, the artifact contains a *minimal independent verifier* (`verify_M.py`, about 100 lines, written directly from Definition 3.4 with no shared code), which re-derives $M(T_b)$ and the complete list of admissible visits by brute force for $b \leq 14$.

**Verified statements.** Each item states the claim, its range, and its log.

1. **Census anchors** (`search_small.log`): $\rho_{\min}(7) = 5$ and $\rho_{\min}(8) = 6$ by full enumeration of $S_7, S_8$ with the Python solvers, independently of the C++ censuses of item 4; see Appendix B.

2. **Optimality of the gadget at small sizes** (`search_small.log`): full enumeration of $S_b$ for $4 \leq b \leq 9$ gives $\min_{\pi \in S_b} M(\pi) = 3$ in every case; no gadget of size $\leq 9$ beats transit number 3.

3. **Exact values** (`endtoend.log`, `bigtable.log`): for $\sigma_{a,b}$:

| $a$ | $b$ | $n$ | $\rho$ (exact) | $2a{+}2b{-}4$ | $4a{+}2b$ | $\rho/n$ |
|---|---|---|---|---|---|---|
| 10 | 8 | 80 | 32 | 32 | 56 | 0.400 |
| 12 | 8 | 96 | 36 | 36 | 64 | 0.375 |
| 12 | 12 | 144 | 44 | 44 | 72 | 0.306 |
| 16 | 12 | 192 | 52 | 52 | 88 | 0.271 |
| 20 | 12 | 240 | 60 | 60 | 104 | 0.250 |
| 16 | 16 | 256 | 60 | 60 | 96 | 0.234 |
| 24 | 16 | 384 | 76 | 76 | 128 | 0.198 |
| 20 | 20 | 400 | 76 | 76 | 120 | 0.190 |
| 30 | 16 | 480 | 88 | 88 | 152 | 0.183 |
| 24 | 20 | 480 | 84 | 84 | 136 | 0.175 |
| 30 | 20 | 600 | 96 | 96 | 160 | 0.160 |
| 36 | 20 | 720 | 108 | 108 | 184 | 0.150 |

Every exact value equals $2a+2b-4$ and sits inside the proven window $[2a{+}2b{-}7,\ 4a{+}2b]$ of (1); the two rows at $n = 480$ (differing in $\rho$) illustrate that $\rho(\sigma_{a,b})$ tracks $a$ and $b$ separately, not just their product.

4. **Exhaustive small-$n$ statistics** (`logs/exh7.txt`–`exh12.txt`, C++ solver): the joint distribution of $(b(\sigma), \rho(\sigma))$ over all of $S_n$ for $7 \le n \le 12$, the source of the $\rho_{\min}$ census of Appendix B and of the deficient-set counts $|D(n)|$ quoted in Section 8; at $n = 12$ this enumerates all 479,001,600 permutations.

5. **The odd gadget and the spectrum** (`verify_oddb.log`, `verify_spectrum.log`): the inventory, highway and crossing claims of Appendix A confirmed uniformly for odd $b \in \{7, 9, \ldots, 21\}$, and $M(T'_b) = 3$ there and at the degenerate $b = 5$; as positive control the same checker reproduces $M(T_b) = 3$ (even $b$) and $M(T_b) = b$ (odd $b$) for the even-gadget family. Separately, exhaustive computation of $M(\pi)$ for every $\pi \in S_b$, $2 \le b \le 8$, yields $\mu(2) = 2$, $\mu(b) = 3$ for $3 \le b \le 8$, and the attained spectra of Remark 7.9 (`verify_spectrum.log`).

Every item is exhaustive or certified within its stated range and reproducible by running the listed scripts.

The $k$-color and matching-content computations behind Proposition 8.1, Observation 8.2, and the exact $f(n)$ values are likewise included in the ancillary artifact: a three-color extension of the search engine with the deficient-set/coset-triangle reduction (`k3.cpp`) and an independent integer-programming certification of all twelve witnesses (`cpsat_rho3.py`); an exact bitmask dynamic program for $\max_\pi \mu(\pi)$ over $G(n, \sigma)$ (`fmu.cpp`; validated against a direct search on all of $S_n$ for $n \le 7$, and against the conversion inequality $\max_\pi \mu \ge \rho$ pointwise on every permutation processed); and, for the $f(10)$–$f(12)$ values, a pruned decision procedure for "$\max_\pi \mu \ge n-1$?" run over the deficient sets $D(10)$–$D(12)$ only (Observation 8.2 covers the rest of $S_n$), validated by exact agreement with the dynamic program on all of $D(10)$ and $D(11)$, with every $\max_\pi \mu = n-2$ exception certified independently by the integer-programming formulation (optimality in all 1120 cases); an integer-programming formulation for $\max_\pi \mu$ used for the structured instances (`fmu_cpsat.py`, optimality certified in all reported cases).

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